

The Physical Celestial Clebsch–Gordan Transform of a Radiative Bose Pair

Direct-integral completion, source-Weyl geometry, and all-label uniqueness on the intrinsic physical source range

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Abstract

The preceding paper shows that the source-aware two-particle celestial Clebsch–Gordan kernel is not fixed-source fiberwise isometric. Here we construct its physical direct-integral framework and determine whether the intrinsic physical source remains uniquely identifiable.

For fixed external labels a, b , the normalized kernel defines angular Clebsch–Gordan rows on a dense ordered separable pair core. If one explicit uniform core estimate holds, these rows extend uniquely to a bounded retained direct-integral transform

$$C_{\text{act}}^{a,b} : \mathcal{H}_{\text{src}} \rightarrow \mathcal{H}_{\text{ang}}^{\text{ret}},$$

with strongly measurable external-label dependence and global assembly over the physical two-particle carrier. This completion theorem is conditional on that uniform estimate.

Independently of the retained completion, we prove unconditional injectivity of the intrinsic cross-ratio transform $T_{\text{sh}}^{a,b}$ and its full signed principal-series continuation $T_{\text{full}}^{a,b}$ on the intrinsic physical source range

$$\mathcal{R}_{\text{phys}}^{a,b} = \text{Ran}\left(\mathcal{D}_{\text{pair}} \xrightarrow{\iota_{a,b}} \mathcal{H}_{\text{src}}\right) : \quad \ker T_{\text{sh}}^{a,b} = \ker T_{\text{full}}^{a,b} = \{0\}$$

for every physical external-label pair.

The proof retains the full unequal-label source character. Source-pair coarea converts physical output nullity into shifted Mellin moments of the original relative-scale source. These moments factor into a source coefficient and an explicitly evaluated spatial beta coefficient. Same-sign modes reduce to a nonzero Gamma quotient; the finite opposite-sign degree-eight sector reduces by Gamma recurrence to nonvanishing quartic factors. A logarithmic-cylinder Fourier-uniqueness theorem then forces the original intrinsic source to vanish.

Thus the source-aware correction is required to realize the physical kernel’s Weyl law, while the corrected transform is injective on $\mathcal{R}_{\text{phys}}^{a,b}$ despite the failure of fixed-source fiberwise isometry. No bounded inverse, uniform lower frame bound, injectivity on all of completed $\mathcal{H}_{\text{src}}$, or injectivity after the additional retained fundamental-sheet projection is claimed.

1 Introduction

The first five papers in this series separate successive mathematical levels of one radiative two-particle problem. Paper 1 constructs a nontruncatable constrained Gaussian sector [1]. Paper 2 identifies the quadratic pair geometry generating its completed even Fock expansion [2]. Paper 3 establishes the Bose/Fock and Poincaré representation of the degree-two sector while keeping physical asymptotic-particle identity as a separate statement [3]. Paper 4 follows the same pair geometry into variational and radiative descendants and computes the associated information loss [4]. Paper 5 then studies the celestial Clebsch–Gordan problem and proves that shadow reflection of the output channel is accompanied by a nontrivial unitary transformation of the fixed-external-label source [5].

The faithfulness question is nontrivial precisely because Paper 5 removes the stronger fixed-source metric picture that would otherwise make information preservation look automatic. In the physical kernel, output shadow reflection does not act on the output alone: it forces a nontrivial source transformation, and the canonical fixed-source transition fails to be fiberwise isometric on a positive-measure spectral set. The present theorem therefore does not reinstate an uncorrected change-of-basis interpretation. It asks whether the *source-aware* transform forced by the kernel remains injective after the fixed-source/isometric shortcut has been invalidated.

Paper 5 hands forward two mathematically distinct structures. The first is a scalar joint source/output representation completion,

$$\mathcal{H}_{\text{joint}}^{\text{fix}}(a, b) \xrightarrow{\simeq_u} \mathcal{H}_{\text{mf}}, \quad (1)$$

where one fundamental principal-series sheet determines the full phase-bearing joint orbit. The second is the angular-valued physical Clebsch–Gordan coefficient

$$C_{J,\nu}^{a,b}[s] \in \mathcal{H}_J^{\text{ang}}, \quad s \in \mathcal{D}_{\text{pair}}, \quad (2)$$

defined first on the dense ordered separable pair core. The distinction is structural: Eq. (1) is a scalar representation theorem on a joint source/output space, whereas Eq. (2) is the angular coefficient produced by the physical kernel.

The present paper asks two linked questions:

How does the physical kernel assemble into a completed celestial transform, and does the intrinsic physical source remain uniquely identifiable from the complete cross-ratio and full signed principal-series celestial data?

For fixed external labels a, b , let $\mathcal{H}_{\text{src}}$ denote the completed fixed-external-label source Hilbert space and let

$$\mathcal{H}_{\text{ang}}^{\text{ret}} = \int_{\mathcal{F}}^{\oplus} \mathcal{H}_J^{\text{ang}} d\mu_{\text{Pl}}(J, \nu) \quad (3)$$

be the retained angular principal-series direct integral. The literal normalized kernel defines an ordered fixed-label presentation-core map

$$C_{\text{direct}}^{a,b} : \mathcal{D}_{\text{pair}} \longrightarrow \mathcal{H}_{\text{ang}}^{\text{ret}}. \quad (4)$$

Its bounded completion is conditional on one explicit estimate:

$$\exists K \geq 0 \quad \|C_{\text{direct}}^{a,b}[s]\| \leq K \|\iota_{a,b}s\| \quad \forall a, b, s \in \mathcal{D}_{\text{pair}}. \quad (5)$$

If Eq. (5) holds, density gives a unique bounded retained transform

$$C_{\text{act}}^{a,b} : \mathcal{H}_{\text{src}} \longrightarrow \mathcal{H}_{\text{ang}}^{\text{ret}}, \quad (6)$$

and the external-label dependence assembles into

$$\mathcal{H}_{\text{phys}}^{(2)} \xrightarrow{U_{\text{src}}} L_{\text{ext}}^2(\mathcal{H}_{\text{src}}) \xrightarrow{C_{\text{ext}}} L_{\text{ext}}^2(\mathcal{H}_{\text{ang}}^{\text{ret}}). \quad (7)$$

This is a conditional completion theorem; the manuscript does not assume Eq. (5) silently.

The direct-integral codomain is essential. A completed L^2 output is an almost-everywhere equivalence class, so arbitrary fixed-label evaluation is not intrinsic. Literal rows remain available on the dense kernel domain through their canonical kernel representatives, while the completed retained transform is correctly represented by a global direct-integral class.

The source-Weyl geometry is treated with the same domain discipline. The kernel-level shadow relation is literal on the regular kernel domain. Separately, the associated source transformation

$$U_{J,\nu}^{\text{src}} : \mathcal{H}_{\text{src}} \xrightarrow{\simeq_u} \mathcal{H}_{\text{src}} \quad (8)$$

is defined on the completed source Hilbert space. A dense source-Weyl-invariant linear domain is obtained by taking the span of all $U_{J,\nu}^{\text{src}}$ -translates of the physical source range. We do not identify arbitrary completed L^2 vectors with literal pointwise kernel rows.

The central unconditional result concerns the intrinsic physical source range

$$\mathcal{R}_{\text{phys}}^{a,b} := \text{Ran} \left(\mathcal{D}_{\text{pair}} \xrightarrow{\iota_{a,b}} \mathcal{H}_{\text{src}} \right). \quad (9)$$

This range is dense in $\mathcal{H}_{\text{src}}$, but it is not assumed closed or complete. The intrinsic cross-ratio transform

$$T_{\text{sh}}^{a,b} : \mathcal{R}_{\text{phys}}^{a,b} \longrightarrow \mathcal{H}_{\text{shape}} \quad (10)$$

and its exact full signed principal-series continuation satisfy

$$\ker T_{\text{sh}}^{a,b} = \ker T_{\text{full}}^{a,b} = \{0\} \quad \forall a, b. \quad (11)$$

Equivalently,

$$T_{\text{sh}}^{a,b} \psi_1 = T_{\text{sh}}^{a,b} \psi_2 \implies \psi_1 = \psi_2, \quad \psi_1, \psi_2 \in \mathcal{R}_{\text{phys}}^{a,b}. \quad (12)$$

The proof is direct and constructive at the level required for uniqueness. Physical nullity is transported to the original ordered source-pair geometry; the unequal-label source character is retained; translated Gaussian observers generate all signed angular harmonics; a shifted Mellin factorization separates the original source coefficient from an explicitly evaluated two-puncture beta coefficient; that coefficient is nonzero for every physical label pair, every integer mode and every real Mellin frequency; and Fourier uniqueness on the logarithmic cylinder recovers the original source.

This resolves the information question left open by Paper 5 without changing its metric conclusion. The source-aware correction is required to realize the physical kernel’s Weyl law, and the corrected transform remains injective:

$$\begin{aligned} &\text{the physical kernel’s Weyl law requires a nontrivial source-aware correction,} \\ &\text{and the corrected physical transform is injective on } \mathcal{R}_{\text{phys}}^{a,b}. \end{aligned} \tag{13}$$

Thus faithfulness here is not inherited from a fixed-source unitary change of basis; it is established after that stronger explanation has failed.

The paper first constructs the literal and retained direct-integral framework, then records the source-Weyl geometry and the intrinsic cross-ratio and full signed principal-series carriers. The central uniqueness argument is developed from explicit conventions and measures, followed by coarea, Gaussian testing, Mellin factorization, beta nonvanishing and logarithmic-cylinder Fourier uniqueness. The concluding sections separate this injectivity theorem from stronger claims about stable inversion and from the additional retained fundamental-sheet projection.

1.1 Scope

The manuscript is noninteracting. It concerns the physical two-particle carrier, the normalized celestial Clebsch–Gordan kernel, direct-integral completion, shadow/source symmetry and the kernel structure of the resulting transform. Scattering operators, action vertices, collinear limits, splitting amplitudes, soft-current algebras, memory and interacting Ward identities require additional dynamical input and are outside the present construction.

1.2 Relation to nearby work

Unitary representation theory of the Lorentz group organizes massless states by little-group data and admits principal-series harmonic analysis [6, 7, 8]. Celestial bases reorganize momentum-space states as conformal primaries on the sphere [10, 11, 12]. Multiparticle celestial states naturally lead to tensor-product decomposition and composite principal-series channels [13]. The reflection $(J, \nu) \mapsto (-J, -\nu)$ is represented by a Knapp–Stein/shadow intertwiner [9]. The present construction keeps the source transformation that accompanies this reflection in the literal two-particle kernel and studies the resulting transform at the completed Hilbert-space level.

1.3 Dictionary to standard celestial conventions

We use

$$\Delta = 1 + i\nu, \quad h = \frac{1 + i\nu + J}{2}, \quad \bar{h} = \frac{1 + i\nu - J}{2}. \tag{14}$$

The reflection

$$(J, \nu) \mapsto (-J, -\nu) \tag{15}$$

is the principal-series shadow transformation

$$(\Delta, J) \mapsto (2 - \Delta, -J), \quad (h, \bar{h}) \mapsto (1 - h, 1 - \bar{h}). \tag{16}$$

For an external graviton label a , we write its helicity as $j_a \in \{+2, -2\}$ and its pre-Mellin Fourier variable as f_a , with

$$\nu_a = 2\pi f_a, \quad \Delta_a = 1 + i\nu_a. \quad (17)$$

The output variable ν is already the principal-series frequency.

Notation	Celestial interpretation	Role in this paper
(J, ν)	output spin and principal-series frequency	label of the angular composite channel
$\Delta = 1 + i\nu$	conformal dimension	output principal-series parameter
$\mathcal{H}_J^{\text{ang}}$	angular principal-series Hilbert fiber	codomain of a literal kernel row
$\mathcal{H}_{\text{ang}}^{\text{ret}}$	retained fundamental-sheet angular direct integral	completed fixed-external-label physical output
$\mathcal{H}_{\text{src}}$	fixed-external-label source Hilbert space	domain of the physical kernel
$U_{J,\nu}^{\text{src}}$	source transition unitary	source action paired with shadow reflection
$\mathcal{S}_{-\nu,J}$	integral-spin shadow/Knapp–Stein map	angular output intertwiner
$\gamma_{a,b}(J, \nu)$	channel phase	unit-modulus coefficient in the Weyl law
$\mathcal{H}_{\text{mf}}$	scalar source-aware completion of Paper 5	representation geometry, distinct from angular output

2 Physical pair source and angular coefficient

2.1 Physical two-particle carrier

Let $\mathcal{H}_{\text{rad}}^{(1)}$ denote the completed one-particle radiative Hilbert space and let $\mathcal{H}_{\text{phys}}^{(2)}$ be the closed Bose-symmetric two-particle sector. The preceding papers establish the physical one-particle and pair carriers and the exact one-particle Poincaré action. The present paper begins from the completed pair space itself.

The celestial analysis preserves the external one-particle provenance. We denote the external multiplicity coordinate by

$$e = (a, b), \quad a = (j_a, f_a), \quad b = (j_b, f_b), \quad (18)$$

with the product measure $d\mu_{\text{ext}}(e)$. There is an exact source disintegration

$$U_{\text{src}} : \mathcal{H}_{\text{phys}}^{(2)} \xrightarrow{\simeq} L_{\text{ext}}^2(\mathcal{H}_{\text{src}}^{\text{sp}}), \quad (19)$$

where $\mathcal{H}_{\text{src}}^{\text{sp}}$ is the completed source spectrum obtained from the fixed-external-label source by unitary cylinder/Fourier coordinates. In particular,

$$\|U_{\text{src}}\Phi\| = \|\Phi\|. \quad (20)$$

2.2 Ordered fixed-label test core and Bose physical carrier

Let $\mathcal{S}_{\text{sep}}^{\text{rad}}$ denote the finite two-helicity separable Schwartz one-particle core. Each element has only the two physical graviton helicity rows, and each row is a finite linear combination of separable frequency–angular profiles with Schwartz frequency dependence and continuous angular dependence. The fixed-label Clebsch–Gordan analysis is performed on the ordered algebraic presentation core

$$\mathcal{D}_{\text{pair}} = \mathcal{S}_{\text{sep}}^{\text{rad}} \otimes_{\text{alg}} \mathcal{S}_{\text{sep}}^{\text{rad}}. \quad (21)$$

Its canonical inclusion has dense range in the completed ordered celestial two-particle carrier.

This ordered presentation core is distinct from the Bose-symmetric physical Hilbert space. On the physical side, let

$$\mathcal{D}_{\text{rad}}^{(2)} = \mathcal{H}_{\text{rad}}^{(1)} \otimes_{\text{alg}} \mathcal{H}_{\text{rad}}^{(1)} \quad (22)$$

be the algebraic tensor square of the physical one-particle carrier. Completion followed by orthogonal projection to the exchange-fixed subspace gives

$$\mathcal{D}_{\text{rad}}^{(2)} \longrightarrow \widehat{\mathcal{H}_{\text{rad}}^{(1)} \otimes \mathcal{H}_{\text{rad}}^{(1)}} \xrightarrow{P_{\text{Bose}}} \mathcal{H}_{\text{phys}}^{(2)}, \quad (23)$$

and this projected algebraic core has dense range in $\mathcal{H}_{\text{phys}}^{(2)}$. Thus the two uses of “pair core” have different roles: $\mathcal{D}_{\text{pair}}$ is the ordered fixed-label celestial presentation on which the literal kernel is evaluated, whereas Eq. (23) supplies the dense Bose-symmetric physical carrier. The exact physical/celestial pair unitary intertwines exchange, so these two descriptions are compatible without identifying the ordered presentation itself with the Bose subspace.

For each external pair (a, b) , the ordered fixed-external-label source map is

$$\iota_{a,b} : \mathcal{D}_{\text{pair}} \longrightarrow \mathcal{H}_{\text{src}}. \quad (24)$$

Its range is dense in $\mathcal{H}_{\text{src}}$. The finite Schwartz/separable structure of Eq. (21) supplies the source envelopes and weighted integrability used in the uniqueness proof; these properties are collected explicitly in Section 11.

For $s \in \mathcal{D}_{\text{pair}}$, the normalized literal Clebsch–Gordan coefficient at output label (J, ν) is

$$C_{J,\nu}^{a,b}[s] \in \mathcal{H}_J^{\text{ang}}. \quad (25)$$

The kernel normalization includes the Plancherel density and the angular normalization fixed in the preceding analysis. The retained fundamental-sheet angular output combines the nonnegative zero-spin half-line with all positive integral spins,

$$\mathcal{F} = \{(0, \nu) : \nu \geq 0\} \cup \{(J, \nu) : J > 0, \nu \in \mathbb{R}\}. \quad (26)$$

The corresponding angular direct integral is

$$\mathcal{H}_{\text{ang}}^{\text{ret}} = \left(\int_{\nu \geq 0}^{\oplus} \mathcal{H}_0^{\text{ang}} d\mu_{\text{Pl}}(0, \nu) \right) \oplus \bigoplus_{J=1}^{\infty} \left(\int_{\mathbb{R}}^{\oplus} \mathcal{H}_J^{\text{ang}} d\mu_{\text{Pl}}(J, \nu) \right). \quad (27)$$

The ordered fixed-label presentation-core transform is

$$C_{\text{direct}}^{a,b}[s] \in \mathcal{H}_{\text{ang}}^{\text{ret}}. \quad (28)$$

Its zero-spin representative and every positive-spin representative agree almost everywhere with the literal normalized kernel row in Eq. (25).

3 Why the completed transform is direct-integral valued

The completed codomain in Eq. (27) is an L^2 direct integral, hence its vectors are defined only up to almost-everywhere equality. Evaluation at a prescribed spectral label is therefore not an intrinsic operation on the completed class. Literal kernel rows remain canonical on $\mathcal{D}_{\text{pair}}$, where the kernel representative is defined before passage to the direct-integral quotient.

Accordingly, a bounded completion

$$C_{\text{act}}^{a,b} : \mathcal{H}_{\text{src}} \longrightarrow \mathcal{H}_{\text{ang}}^{\text{ret}}$$

is represented by a global direct-integral class together with almost-everywhere agreement with the core kernel rows. A pointwise bounded operator field would require additional regularity beyond Hilbert completion itself.

4 The single boundedness hypothesis

4.1 Uniform core estimate

The completed transform requires one Hilbert-space estimate. We isolate it in its weakest direct form.

Uniform core bound. There exists $K \geq 0$ such that for every external pair a, b and every $s \in \mathcal{D}_{\text{pair}}$,

$$\|C_{\text{direct}}^{a,b}[s]\| \leq K \|\iota_{a,b}s\|. \quad (29)$$

The estimate in Eq. (29) is equivalent to the uniform finite-window energy condition that appears naturally when the spin sum and frequency intervals are introduced before the global limit.

No exact norm identity is included in Eq. (29). In particular,

$$\|C_{\text{direct}}^{a,b}[s]\| = \|\iota_{a,b}s\| \quad (30)$$

is not assumed. Paper 5 isolates a stronger literal unit-Gram problem and proves that the full requested fixed-source identification fails through an explicit transition coefficient on a positive-measure frequency interval [5].

4.2 Unique fixed-external-label completion

Assume Eq. (29). Since $\iota_{a,b}(\mathcal{D}_{\text{pair}})$ is dense in $\mathcal{H}_{\text{src}}$, the map in Eq. (28) extends uniquely to

$$C_{\text{act}}^{a,b} : \mathcal{H}_{\text{src}} \longrightarrow \mathcal{H}_{\text{ang}}^{\text{ret}} \quad (31)$$

with

$$\|C_{\text{act}}^{a,b}\psi\| \leq K\|\psi\|. \quad (32)$$

The extension obeys

$$C_{\text{act}}^{a,b}(\iota_{a,b}s) = C_{\text{direct}}^{a,b}[s] \quad (33)$$

for every ordered fixed-label presentation-core source.

4.3 The stronger coherent problem

The full source-side construction contains a coherent object before retained-sheet restriction. Its natural domain is the image of the normalized phase-incidence isometry

$$I_{a,b} : \mathcal{H}_{\text{src}} \xrightarrow{\cong} \mathcal{R}_{a,b} \subset \mathcal{H}_{\text{inc}}. \quad (34)$$

The image $\mathcal{R}_{a,b}$ is closed, and the image of the ordered fixed-label presentation core is dense in $\mathcal{R}_{a,b}$. On the corresponding dense ordered fixed-label incidence range there is a linear coherent operator

$$T_{a,b}^0 : D_{a,b} \subset \mathcal{R}_{a,b} \longrightarrow \mathcal{H}_{\text{coh}}. \quad (35)$$

The exact deweighting formula contains the incidence half-density and the Rühl-scale norm in the combination

$$h(b, z) \|q_{\text{Rühl}}(b, z)\| = 1 \quad (36)$$

on the regular physical chart. Thus the apparent deweighting growth cannot be interpreted as an ambient-space obstruction without respecting the physical incidence range.

The remaining full-coherent question has the exact operator-theoretic form

$$\begin{aligned} \exists K_{\text{coh}} \geq 0 : \|T_{a,b}^0 x\| \leq K_{\text{coh}} \|x\| \quad \text{uniformly in } a, b \\ \iff \text{there exists a unique uniformly bounded extension } T_{a,b} : \mathcal{R}_{a,b} \rightarrow \mathcal{H}_{\text{coh}}. \end{aligned} \quad (37)$$

The full coherent bound implies Eq. (29). The converse is not supplied by retained-sheet control. The omitted signed rows are reconstructed by the source-twisted law only after the source itself has been changed.

The present formal development proves neither a finite universal K_{coh} nor a normalized sequence on the physical range for which $\|T^0 x_n\| \rightarrow \infty$. Equation (37) is therefore the terminal full-coherent analytic question, while Eq. (29) is the sole assumption needed for the retained physical transform developed below.

5 External-label measurability and global assembly

The physical pair carries external helicity and frequency data. A global celestial transform therefore requires a measurable field of fixed-external-label operators over the external multiplicity coordinate.

5.1 A common dense source core

There is a label-independent dense source core

$$\mathcal{D}_{\text{src}}^{\text{sp}} \subset \mathcal{H}_{\text{src}}^{\text{sp}} \quad (38)$$

obtained by transporting the continuous angular pair core through the exact source spectral unitary. Its range is dense:

$$\overline{\mathcal{D}_{\text{src}}^{\text{sp}}} = \mathcal{H}_{\text{src}}^{\text{sp}}. \quad (39)$$

For every fixed core vector $u \in \mathcal{D}_{\text{src}}^{\text{sp}}$, the external orbit

$$e \longmapsto C_{\text{act}}^e u \quad (40)$$

is strongly measurable. The proof uses the jointly measurable literal kernel and scalar matrix coefficients against a countable dense target family, followed by the Hilbert-space Pettis criterion.

Uniform boundedness then extends Eq. (40) from the dense source core to every fixed $u \in \mathcal{H}_{\text{src}}^{\text{sp}}$. Finally, simple-function approximation gives strong measurability for arbitrary Bochner sections

$$f \in L_{\text{ext}}^2(\mathcal{H}_{\text{src}}^{\text{sp}}), \quad e \longmapsto C_{\text{act}}^e f(e). \quad (41)$$

Thus external measurability is a theorem once Eq. (29) is supplied; it is not an additional assumption.

5.2 External direct-integral operator

The fiberwise maps therefore assemble to a bounded operator

$$C_{\text{ext}} : L_{\text{ext}}^2(\mathcal{H}_{\text{src}}^{\text{sp}}) \longrightarrow L_{\text{ext}}^2(\mathcal{H}_{\text{ang}}^{\text{ret}}). \quad (42)$$

Its representative satisfies

$$(C_{\text{ext}} f)(e) = C_{\text{act}}^e f(e) \quad \text{for a.e. } e, \quad (43)$$

and

$$\|C_{\text{ext}} f\| \leq K \|f\|. \quad (44)$$

Composing with the physical source disintegration in Eq. (19) gives the physical celestial transform

$$C_{\text{phys}} := C_{\text{ext}} U_{\text{src}} : \mathcal{H}_{\text{phys}}^{(2)} \longrightarrow L_{\text{ext}}^2(\mathcal{H}_{\text{ang}}^{\text{ret}}). \quad (45)$$

The norm estimate is

$$\|C_{\text{phys}} \Phi\| \leq K \|\Phi\|. \quad (46)$$

For almost every external coordinate,

$$(C_{\text{phys}} \Phi)(e) = C_{\text{act}}^e ((U_{\text{src}} \Phi)(e)). \quad (47)$$

Equations (45)–(47) are the completed physical two-particle celestial transform established in this paper.

6 Source-Weyl geometry of the physical kernel

The source-Weyl relation has two levels that must be kept separate.

6.1 Kernel-level identity and completed source action

On the regular three-point kernel domain, shadow reflection of the output label $(J, \nu) \mapsto (-J, -\nu)$ is accompanied by multiplication of the input-input chord by the unit character

$$\Xi_{J,\nu}(z) = \mathbf{N}(z; 2i\nu, 2J), \quad z \neq 0, \quad (48)$$

For a regular three-point configuration with output point z and input points z_1, z_2 , the literal distributional kernel obeys

$$\boxed{K_{J,\nu}^{a,b}(z; z_1, z_2) \Xi_{J,\nu}(z - z_1) = K_{-J,-\nu}^{a,b}(z; z_1, z_2) \Xi_{J,\nu}(z_1 - z_2) \Xi_{J,\nu}(z_2 - z).} \quad (49)$$

This is the kernel-level identity before passage to the direct-integral quotient. The factor on the input chord $z_1 - z_2$ is the source transformation; the remaining factors belong to the angular output-shadow transport.

Multiplication by $\Xi_{J,\nu}(z_1 - z_2)$ has unit modulus off the pair diagonal. It therefore defines a unitary source action

$$\boxed{U_{J,\nu}^{\text{src}} : \mathcal{H}_{\text{src}} \xrightarrow{\simeq_u} \mathcal{H}_{\text{src}},} \quad (50)$$

whose inverse is $U_{-J,-\nu}^{\text{src}}$. For a completed source represented by $\psi(z_1, z_2)$,

$$(U_{J,\nu}^{\text{src}}\psi)(z_1, z_2) = \Xi_{J,\nu}(z_1 - z_2)\psi(z_1, z_2) \quad \text{a.e.} \quad (51)$$

The pair diagonal is measure zero and does not affect the L^2 class.

The literal row notation $C_{J,\nu}^{a,b}[s]$ is reserved for $s \in \mathcal{D}_{\text{pair}}$, where the kernel representative is defined before quotienting. We do not write $C_{J,\nu}^{a,b}[\psi]$ for an arbitrary completed $\psi \in \mathcal{H}_{\text{src}}$.

To formulate source transport on a domain stable under every $U_{J,\nu}^{\text{src}}$, define

$$\mathcal{D}_W^{a,b} := \text{span}_{\mathbb{C}} \left\{ U_{J,\nu}^{\text{src}}\psi : \psi \in \mathcal{R}_{\text{phys}}^{a,b}, J \in \mathbb{Z}, \nu \in \mathbb{R} \right\}. \quad (52)$$

Because $\mathcal{R}_{\text{phys}}^{a,b}$ is dense in $\mathcal{H}_{\text{src}}$, $\mathcal{D}_W^{a,b}$ is dense. It is invariant under every source-Weyl unitary and under the inverse action. This gives a domain-stable source representation without asserting that the original core-generated range itself is invariant under every label.

Remark on the completed source orbit. For every $\psi \in \mathcal{H}_{\text{src}}$, the measurable orbit

$$(J, \nu) \mapsto U_{J,\nu}^{\text{src}}\psi$$

has constant norm $\|\psi\|$. Hence a nonzero orbit is not itself a global L^2 section over the infinite-measure signed-label space $\mathbb{Z} \times \mathbb{R}$, although its restriction to every finite-measure label window is square integrable. This keeps the measurable unitary source relation distinct from the globally square-integrable joint source/output completion of Paper 5.

7 Retained and signed representations

Paper 5 constructs the exact joint source/output spectral involution associated with the source-Weyl transformation described in Section 6. In source spectral coordinates the label reflection and source translation form a measure-preserving involution, and multiplication by the channel phase produces a unitary symmetry of the joint scalar L^2 carrier [5].

The corresponding fixed sector admits a two-sided unitary reduction to the retained fundamental sheet:

$$\mathcal{H}_{\text{joint}}^{\text{fix}}(a, b) \xrightarrow{\sim_u} \mathcal{H}_{\text{mf}}. \quad (53)$$

The two inverse identities are

$$\mathcal{A}_{a,b} \mathcal{B}_{a,b} = \mathbf{1}_{\mathcal{H}_{\text{mf}}}, \quad \mathcal{B}_{a,b} \mathcal{A}_{a,b} = \mathbf{1}_{\mathcal{H}_{\text{joint}}^{\text{fix}}}. \quad (54)$$

At the kernel level, omitted signed rows are reconstructed from retained rows together with the source transformation and the angular shadow intertwiner described in Section 6. Abstractly, for bounded source-aware operator families satisfying the same exact law, restriction to one fundamental sheet is injective and reconstruction is two-sided.

The completed physical angular transform and the scalar joint completion therefore occupy complementary roles:

$$\begin{aligned} \mathcal{H}_{\text{ang}}^{\text{ret}} &: \text{angular-valued physical output,} \\ \mathcal{H}_{\text{mf}} &: \text{scalar source-aware Weyl representation completion.} \end{aligned} \quad (55)$$

The shared structure is the same phase, shadow reflection, source transition and retained/full reconstruction law. No Hilbert-space identification between the two carriers is required.

8 Intrinsic cross-ratio and full signed principal-series transforms

The retained direct-integral map is obtained after an additional fundamental-sheet projection. The uniqueness theorem is proved earlier in the factorization, on the intrinsic physical source range $\mathcal{R}_{\text{phys}}^{a,b}$.

For a core source $s \in \mathcal{D}_{\text{pair}}$, let

$$\Psi_s^{a,b}(z_1, z_2)$$

be the source-coordinate ordered-pair representative. For $Z_{\text{out}} \in \mathbb{C}$, $q \in \mathbb{C}^\times$, and $u \in \mathbb{C} \setminus \{0, -1\}$,

$$z_1 = Z_{\text{out}} + \frac{q}{1+u}, \quad z_2 = Z_{\text{out}} + \frac{q}{u}, \quad \lambda = -\frac{u(1+u)}{q}. \quad (56)$$

Let

$$\eta_{a,b}(\lambda, u) = \frac{B_{a,b}(\lambda, u)}{|B_{a,b}(\lambda, u)|}$$

be the normalized physical baseline phase on the regular locus. The phase-corrected source fiber is

$$\mathcal{J}_{a,b,s}(Z_{\text{out}}, q; u) = \eta_{a,b}(\lambda, u) \Psi_s^{a,b}(z_1, z_2). \quad (57)$$

With the cross-ratio measure defined in Eq. (189), the coherent one-shape amplitude is

$$\Phi_s^{a,b}(Z_{\text{out}}, q) = \int_{\mathbb{C}} \mathcal{I}_{a,b,s}(Z_{\text{out}}, q; u) d\mu_{\text{cr}}(u). \quad (58)$$

The one-shape output carrier is

$$\mathcal{H}_{\text{shape}} = L^2\left(\mathbb{C}_{Z_{\text{out}}} \times \mathbb{C}_q^\times, d\mu_{\text{cel}}(Z_{\text{out}}) d^2 q\right), \quad (59)$$

where $d\mu_{\text{cel}}$ is the north-chart round-sphere measure. Passage from a core representative to physical source equality defines

$$T_{\text{sh}}^{a,b} : \mathcal{R}_{\text{phys}}^{a,b} \longrightarrow \mathcal{H}_{\text{shape}}. \quad (60)$$

For a core representative s ,

$$T_{\text{sh}}^{a,b}(\iota_{a,b}s) = [\mathcal{N}_{\text{CG}} \Phi_s^{a,b}]_{L^2}, \quad \mathcal{N}_{\text{CG}} \neq 0. \quad (61)$$

Hence $T_{\text{sh}}^{a,b}(\iota_{a,b}s) = 0$ exactly when $\Phi_s^{a,b} = 0$ almost everywhere with respect to the shape measure.

The full signed principal-series output is obtained by a unitary post-shape chain. At the carrier level,

$$\begin{aligned} \mathcal{H}_{\text{shape}} &\xrightarrow{U_{\text{geom}}} \mathcal{H}_{\text{flat log}} \xrightarrow{U_{\text{dens}}} \mathcal{H}_{\text{flat cyl}} \xrightarrow{U_{\text{FM}}} \mathcal{H}_{\text{cyl spec}} \\ &\xrightarrow{U_{\text{freq}}} \mathcal{H}_{\text{phys freq}} \xrightarrow{U_{\text{decl}}} \mathcal{H}_{\text{signed}} \xrightarrow{U_{\text{sh}}} \mathcal{H}_{\text{full}}. \end{aligned} \quad (62)$$

The factors are, respectively: the scale half-density/logarithmic-cylinder unitary, output-plane density flattening, cylinder Fourier–Mellin Plancherel transform, physical-frequency reparametrization, declared-output convention transport, and signed-label shadow convention. Each arrow is unitary. Define

$$U_{\text{post}} = U_{\text{sh}} U_{\text{decl}} U_{\text{freq}} U_{\text{FM}} U_{\text{dens}} U_{\text{geom}}. \quad (63)$$

Then

$$T_{\text{full}}^{a,b} = U_{\text{post}} T_{\text{sh}}^{a,b}, \quad \|U_{\text{post}} v\| = \|v\|, \quad (64)$$

so

$$\ker T_{\text{full}}^{a,b} = \ker T_{\text{sh}}^{a,b}. \quad (65)$$

The retained transform contains an additional projection after the full-signed representation. Its norm is controlled by the shape output,

$$\|T_{\text{ret}}^{a,b} \psi\| \leq \|T_{\text{sh}}^{a,b} \psi\|, \quad (66)$$

but the reverse estimate needed to infer retained-sector injectivity is not established.

9 Frame and observable quotient of the retained completion

Assume the uniform core estimate in Eq. (29), so that the retained bounded transform

$$C_{\text{act}}^{a,b} : \mathcal{H}_{\text{src}} \rightarrow \mathcal{H}_{\text{ang}}^{\text{ret}}$$

exists. Define

$$F^{a,b} = (C_{\text{act}}^{a,b})^\dagger C_{\text{act}}^{a,b}. \quad (67)$$

Standard Hilbert-space identities give

$$F^{a,b} \geq 0, \quad F^{a,b} = (F^{a,b})^\dagger, \quad \|F^{a,b}\| = \|C_{\text{act}}^{a,b}\|^2, \quad (68)$$

and

$$\ker F^{a,b} = \ker C_{\text{act}}^{a,b}, \quad (\text{Ran } F^{a,b})^\perp = \ker C_{\text{act}}^{a,b}. \quad (69)$$

Thus the observational equivalence relation for the retained transform is

$$\psi \sim_{\text{ret}} \phi \iff C_{\text{act}}^{a,b}\psi = C_{\text{act}}^{a,b}\phi \iff \psi - \phi \in \ker C_{\text{act}}^{a,b}. \quad (70)$$

The quotient

$$\mathcal{Q}_{\text{ret}}^{a,b} = \mathcal{H}_{\text{src}} / \ker C_{\text{act}}^{a,b} \quad (71)$$

carries an injective induced retained transform. Equivalently, the restriction to

$$(\ker C_{\text{act}}^{a,b})^\perp$$

is injective.

These statements describe exactly the observational quotient after the retained fundamental-sheet projection. They do not assert that $\ker C_{\text{act}}^{a,b} = 0$. The all-label uniqueness theorem proved below concerns $T_{\text{sh}}^{a,b}$ and $T_{\text{full}}^{a,b}$ on $\mathcal{R}_{\text{phys}}^{a,b}$, before that additional retained projection.

10 No source-blind equivariant identification

An earlier angular coefficient realization uses a fixed-output carrier in which the independent source-spectral action is absent. The comparison with the source-aware construction is most cleanly expressed representation-theoretically.

Let X carry the identity action and let Y carry an action T . Suppose

$$B : X \rightarrow Y \quad (72)$$

is surjective and intertwines the two actions:

$$T(Bx) = Bx \quad \forall x \in X. \quad (73)$$

Surjectivity gives, for every $y = Bx$,

$$Ty = T(Bx) = Bx = y, \quad (74)$$

so

$$T = \mathbf{1}_Y. \quad (75)$$

The source action is nontrivial. In particular, there is a completed Gaussian source ψ_G such that

$$U_{1,0}^{\text{src}}\psi_G \neq \psi_G. \quad (76)$$

Therefore no surjective comparison map can intertwine the declared source-blind identity action with the physical source transformation.

This result concerns the specified representation structures. It does not assert that an angular Hilbert space cannot support some other nontrivial representation of the source group. The theorem isolates the obstruction to identifying the historical fixed-output realization with the source-aware representation while preserving the source action.

11 Physical source core, analytic domain and descent

The injectivity proof uses only source vectors in

$$\mathcal{R}_{\text{phys}}^{a,b} = \text{Ran} \left(\mathcal{D}_{\text{pair}} \xrightarrow{\iota_{a,b}} \mathcal{H}_{\text{src}} \right).$$

This section records the analytic properties inherited from the finite separable Schwartz core.

11.1 One-particle affine profile, tensor factorization and one-chord domain

Define the positive north-chart half-density

$$w_{\text{cel}}(z) = \frac{2}{1 + |z|^2}. \quad (77)$$

For an external label a and a one-particle core profile $f \in \mathcal{S}_{\text{sep}}^{\text{rad}}$, let

$$A_{a,f} : S^2 \longrightarrow \mathbb{C} \quad (78)$$

denote its continuous angular representative at the selected external label. The affine source profile is exactly

$$\psi_{a,f}(z) = w_{\text{cel}}(z) A_{a,f}(\hat{n}(z)), \quad (79)$$

where $\hat{n}(z)$ is the stereographic point on the celestial sphere. Consequently

$$|\psi_{a,f}(z)| \leq \|A_{a,f}\|_{\infty} w_{\text{cel}}(z). \quad (80)$$

For a pure tensor $s = f \otimes g \in \mathcal{D}_{\text{pair}}$, the source-coordinate ordered-pair representative factorizes exactly:

$$\Psi_{f \otimes g}^{a,b}(z_1, z_2) = \psi_{a,f}(z_1) \psi_{b,g}(z_2). \quad (81)$$

Combining Eqs. (80) and (81) gives

$$|\Psi_{f \otimes g}^{a,b}(z_1, z_2)| \leq \|A_{a,f}\|_{\infty} \|A_{b,g}\|_{\infty} w_{\text{cel}}(z_1) w_{\text{cel}}(z_2). \quad (82)$$

Algebraic tensor induction and the triangle inequality therefore give, for every $s \in \mathcal{D}_{\text{pair}}$, a finite constant $C_s \geq 0$ such that

$$|\Psi_s^{a,b}(z_1, z_2)| \leq C_s w_{\text{cel}}(z_1) w_{\text{cel}}(z_2). \quad (83)$$

The one-chord majorant

$$\mathbf{m}_1(z_1, z_2) = w_{\text{cel}}(z_1) w_{\text{cel}}(z_2) |z_1 - z_2|^{-1} \quad (84)$$

is integrable on $\mathbb{C}^2$. It is the stereographic pullback of the integrable spherical three-chord inverse-distance majorant with its third vertex fixed at the north pole. Thus

$$\int_{\mathbb{C}^2} \mathbf{m}_1(z_1, z_2) d^2 z_1 d^2 z_2 < \infty. \quad (85)$$

Since dephasing has unit modulus off the null diagonal, Eqs. (83)–(85) imply

$$\int_{\mathbb{C}^2} \frac{|\tilde{\Psi}_s^{a,b}(z_1, z_2)|}{|z_1 - z_2|} d^2 z_1 d^2 z_2 < \infty. \quad (86)$$

Introduce once and for all the center/chord coordinates

$$X = z_2, \quad t = z_1 - z_2, \quad (z_1, z_2) = (X + t, X), \quad (87)$$

whose real Jacobian is one:

$$d^2 z_1 d^2 z_2 = d^2 X d^2 t. \quad (88)$$

Define

$$G_{a,b,s}(X, t) = \tilde{\Psi}_s^{a,b}(X + t, X) \frac{\chi_{a,b}^{\text{in}}(t)}{|t|}. \quad (89)$$

Then

$$G_{a,b,s} \in L^1(\mathbb{C}_X \times \mathbb{C}_t). \quad (90)$$

In particular, for almost every X ,

$$G_{a,b,s}(X, \cdot) \in L^1(\mathbb{C}_t). \quad (91)$$

The same finite-core envelope also gives the inverse-radius bound. Since $w_{\text{cel}}(z) \leq 2$,

$$|G_{a,b,s}(X, t)| \leq \frac{C'_s}{|t|} \quad (t \neq 0) \quad (92)$$

for some $C'_s \geq 0$, uniformly in X .

11.2 Fractional multiplicative moments

Recall

$$H_{a,b,s}(X, t) = |t|^2 G_{a,b,s}(X, t), \quad d\mu_{\times}(t) = \frac{d^2 t}{|t|^2}.$$

For every source base point X satisfying Eq. (91) and every

$$0 \leq \alpha < 1,$$

we have

$$\int_{\mathbb{C}^{\times}} |t|^{-\alpha} |H_{a,b,s}(X, t)| d\mu_{\times}(t) = \int_{\mathbb{C}} |t|^{-\alpha} |G_{a,b,s}(X, t)| d^2 t. \quad (93)$$

Split the last integral into $|t| < 1$ and $|t| \geq 1$. On the unit disk, Eq. (92) gives

$$\int_{|t| < 1} |t|^{-\alpha} |G(X, t)| d^2 t \leq 2\pi C'_s \int_0^1 r^{-\alpha} dr < \infty, \quad (94)$$

while on $|t| \geq 1$,

$$|t|^{-\alpha} \leq 1$$

and Eq. (91) supplies the tail. Hence one conull set of source base points supports the entire subcritical strip:

$$\int_{\mathbb{C}^{\times}} |t|^{-\alpha} |H_{a,b,s}(X, t)| d\mu_{\times}(t) < \infty \quad \forall 0 \leq \alpha < 1. \quad (95)$$

The Mellin line used later requires only

$$\int_{\mathbb{C}^{\times}} |t|^{-1/2} |H_{a,b,s}(X, t)| d\mu_{\times}(t) < \infty. \quad (96)$$

Proposition (physical source-domain theorem). For every $s \in \mathcal{D}_{\text{pair}}$ and fixed a, b , the dephased source has the global one-chord domain in Eq. (86); equivalently $G_{a,b,s} \in L^1(\mathbb{C}^2)$. On one conull set of source base points, the Haar-normalized source satisfies every fractional moment in Eq. (95), including the half-power moment in Eq. (96). These are consequences of the ordered fixed-label presentation core itself and are not additional hypotheses of the injectivity theorem.

11.3 Completed coherent shape state and descent to physical source equality

For $s \in \mathcal{D}_{\text{pair}}$, write

$$\mathcal{J}_s^{a,b}(Z_{\text{out}}, q; u) = \eta_{a,b}(\lambda(q, u), u) \Psi_s^{a,b}\left(Z_{\text{out}} + \frac{q}{1+u}, Z_{\text{out}} + \frac{q}{u}\right) \quad (97)$$

for the phase-corrected one-shape source fiber, so that

$$\Phi_s^{a,b}(Z_{\text{out}}, q) = \int_{\mathbb{C}} \mathcal{J}_s^{a,b}(Z_{\text{out}}, q; u) d\mu_{\text{cr}}(u). \quad (98)$$

11.3.1 The doubled-cross-ratio polarized pairing

For $s_1, s_2 \in \mathcal{D}_{\text{pair}}$, define the doubled phase-twisted field

$$\mathcal{P}_{a,b}[s_1, s_2](Z_{\text{out}}, q; u, v) = \overline{\mathcal{J}_{s_1}^{a,b}(Z_{\text{out}}, q; u)} \mathcal{J}_{s_2}^{a,b}(Z_{\text{out}}, q; v). \quad (99)$$

The physical doubled-cross-ratio pairing is

$$\begin{aligned} \Omega_{\text{coh}}^{a,b}(s_1, s_2) &= |\mathcal{N}_{\text{CG}}|^2 \int_{\mathbb{C}_{Z_{\text{out}}}} \int_{\mathbb{C}_q^\times} \int_{\mathbb{C}_u} \int_{\mathbb{C}_v} \mathcal{P}_{a,b}[s_1, s_2](Z_{\text{out}}, q; u, v) \\ &\quad \times d\mu_{\text{cr}}(u) d\mu_{\text{cr}}(v) d^2q d\mu_{\text{cel}}(Z_{\text{out}}). \end{aligned} \quad (100)$$

Incidence scale and shape measures. The physical scale and incidence-shape measures used in this section are

$$d\mu_{\text{scale}}(q) = \mathbf{1}_{\mathbb{C}^\times}(q) |q|^2 d^2q, \quad (101)$$

and

$$d\mu_{\text{sh}}(u) = \mathbf{1}_{\mathbb{C} \setminus \{0, -1\}}(u) \frac{d^2u}{|u|^4 |1+u|^4} =: d_{\text{sh}}(u) d^2u. \quad (102)$$

The coherent cross-ratio measure used after transition-envelope cancellation remains the distinct measure

$$d\mu_{\text{cr}}(u) = \mathbf{1}_{\mathbb{C} \setminus \{0, -1\}}(u) \frac{d^2u}{|u|^2 |1+u|^2},$$

so $d\mu_{\text{sh}}$ and $d\mu_{\text{cr}}$ are not identified.

Proposition (absolute doubled-coherent integrability). For every $s_1, s_2 \in \mathcal{D}_{\text{pair}}$,

$$\mathcal{P}_{a,b}[s_1, s_2] \in L^1\left(d\mu_{\text{cel}}(Z_{\text{out}}) d^2q d\mu_{\text{cr}}(u) d\mu_{\text{cr}}(v)\right). \quad (103)$$

We prove this from the physical incidence majorants already constructed above, rather than importing an external kernel theorem.

For a core source s , define the shape-integrated incidence modulus

$$\mathcal{M}_s(Z, q) = \int_{\mathbb{C}} \left| \mathcal{R}_s^{a,b}(Z; q, u) \right| d\mu_{\text{sh}}(u), \quad (104)$$

where $\mathcal{R}_s^{a,b}$ is the regulator-free source-side incidence function defined explicitly in Eq. (155). In logarithmic scale coordinates

$$q = e^{r+i\theta},$$

put

$$\widetilde{\mathcal{M}}_s(Z; r, \theta) = e^{4r} \mathcal{M}_s(Z, e^{r+i\theta}). \quad (105)$$

Choose a finite algebraic presentation

$$s = \sum_{k=1}^{N_s} c_k f_k \otimes g_k, \quad f_k, g_k \in \mathcal{S}_{\text{sep}}^{\text{rad}}, \quad (106)$$

and define

$$S_s := \sum_{k=1}^{N_s} |c_k| \|A_{a,f_k}\|_{\infty} \|A_{b,g_k}\|_{\infty} < \infty. \quad (107)$$

By the pure-tensor factorization in Eq. (81), S_s is a valid uniform bound for the unweighted regular source amplitude. No canonical choice of finite presentation is required; any such bound is sufficient for the domination argument. To make the fixed-output estimate explicit, define

$$\mathfrak{G}_Z(z_1, z_2) := w_{\text{cel}}(z_1) w_{\text{cel}}(z_2) d_{\text{aff}}(z_1, z_2)^{-1} d_{\text{aff}}(z_2, Z)^{-1} d_{\text{aff}}(Z, z_1)^{-1}, \quad (108)$$

with the inverse-distance factors totalized to zero on their diagonals. On a regular three-point configuration, define the source-coordinate raw incidence integrand by

$$\mathcal{R}_{s,\text{raw}}^{a,b}(z_1, z_2; Z) := \Psi_s^{a,b}(z_1, z_2) K_{0,0}^{a,b}(Z; z_1, z_2), \quad (109)$$

and set it to zero on the null exceptional set. Here $K_{0,0}^{a,b}$ denotes the fixed-external-label kernel appearing in the source-Weyl relation of Section 6. On the regular locus its modulus is exactly

$$\left| K_{0,0}^{a,b}(Z; z_1, z_2) \right| = |\mathcal{N}_{\text{CG}}| |z_1 - z_2|^{-1} |z_2 - Z|^{-1} |Z - z_1|^{-1}. \quad (110)$$

All label dependence on this row is therefore confined to unit phases and the fixed normalization. Combining Eq. (110) with the affine source envelope gives the source-side raw incidence bound

$$\left| \mathcal{R}_{s,\text{raw}}^{a,b}(z_1, z_2; Z) \right| \leq S_s |\mathcal{N}_{\text{CG}}| \mathfrak{G}_Z(z_1, z_2). \quad (111)$$

Under stereographic transport, $\mathfrak{G}_Z$ is the product celestial area density times one output half-density and the spherical three-chord inverse-distance kernel. Hence

$$\int_{\mathbb{C}^2} \mathfrak{G}_Z(z_1, z_2) d^2z_1 d^2z_2 = w_{\text{cel}}(Z) C_{3\text{ch}}, \quad (112)$$

where, with inverse chord distances totalized to zero on the corresponding diagonals,

$$\mathbf{m}_{3\text{ch}}(N; \omega_1, \omega_2) = d_{S^2}(\omega_1, \omega_2)^{-1} d_{S^2}(\omega_2, N)^{-1} d_{S^2}(N, \omega_1)^{-1}, \quad (113)$$

and

$$C_{3\text{ch}} = \int_{S^2 \times S^2} \mathbf{m}_{3\text{ch}}(N; \omega_1, \omega_2) d\Omega_1 d\Omega_2 < \infty. \quad (114)$$

The finiteness of $C_{3\text{ch}}$ is the spherical three-chord integrability used in the one-chord source-domain proof.

The exact incidence and log-scale changes of variables preserve the corresponding measures, so

$$\begin{aligned} \int_{\mathbb{R} \times (-\pi, \pi)} \widetilde{\mathcal{M}}_s(Z; r, \theta) dr d\theta &= \int_{\text{incidence fiber}} |\mathcal{R}_s^{a,b}(Z; \cdot)| d\mu_{\text{inc}} \\ &= \int_{\mathbb{C}^2} |\mathcal{R}_{s,\text{raw}}^{a,b}(z_1, z_2; Z)| d^2 z_1 d^2 z_2 \\ &\leq S_s |\mathcal{N}_{\text{CG}}| C_{3\text{ch}} w_{\text{cel}}(Z). \end{aligned} \quad (115)$$

Since

$$\int_{\mathbb{C}} w_{\text{cel}}(Z) d\mu_{\text{cel}}(Z) = \int_{\mathbb{C}} \frac{8 d^2 Z}{(1 + |Z|^2)^3} = 4\pi, \quad (116)$$

Eq. (115) implies

$$\widetilde{\mathcal{M}}_s \in L^1(d\mu_{\text{cel}}(Z) dr d\theta). \quad (117)$$

For the uniform pointwise estimate, put

$$Q(Z, q; u) = |u|^2 + |Zu + q|^2 \quad (118)$$

and define the positive physical marginal

$$p_{Z,q}(u) = \frac{4|q|}{Q(Z, q; u) Q(Z, q; 1 + u)}. \quad (119)$$

After inserting the shape density and the logarithmic scale density, the source-side incidence norm satisfies

$$|q|^4 d_{\text{sh}}(u) |\mathcal{R}_s^{a,b}(Z; q, u)| \leq S_s |\mathcal{N}_{\text{CG}}| p_{Z,q}(u) d^2 u. \quad (120)$$

Completing the square gives

$$Q(Z, q; u) = (1 + |Z|^2) (|u + c_{Z,q}|^2 + \sigma_{Z,q}^2), \quad (121)$$

with

$$c_{Z,q} = \frac{\bar{Z}q}{1 + |Z|^2}, \quad \sigma_{Z,q} = \frac{|q|}{1 + |Z|^2}.$$

Hence

$$p_{Z,q}(u) = \frac{4}{1 + |Z|^2} \frac{\sigma_{Z,q}}{(|u + c_{Z,q}|^2 + \sigma_{Z,q}^2)(|1 + u + c_{Z,q}|^2 + \sigma_{Z,q}^2)}. \quad (122)$$

Define

$$C_{3/2} = \int_{\mathbb{C}} |u|^{-3/2} |1 + u|^{-3/2} d^2 u < \infty. \quad (123)$$

The finiteness is elementary: near $u = 0$ and $u = -1$ the radial integrand is $O(r^{-1/2}) dr$, while at infinity it is $O(r^{-2}) dr$. The exact domination used here is, for every $\sigma \geq 0$,

$$\frac{\sigma}{(|u|^2 + \sigma^2)(|1 + u|^2 + \sigma^2)} \leq |u|^{-3/2} |1 + u|^{-3/2} \quad \text{for a.e. } u \in \mathbb{C}. \quad (124)$$

For $\sigma > 0$, write the left side as the product

$$\frac{\sqrt{\sigma}}{|u|^2 + \sigma^2} \frac{\sqrt{\sigma}}{|1 + u|^2 + \sigma^2}$$

and use

$$\frac{\sqrt{\sigma}}{r^2 + \sigma^2} \leq r^{-3/2} \quad (r > 0);$$

the case $\sigma = 0$ is immediate. Translation invariance and

$$\frac{4}{1 + |Z|^2} \leq 4$$

therefore give

$$\int_{\mathbb{C}} p_{Z,q}(u) d^2 u \leq 4C_{3/2} \quad (125)$$

uniformly in Z, q . Integrating Eq. (120) in u yields

$$0 \leq \widetilde{\mathcal{M}}_s(Z; r, \theta) \leq B_s := S_s |\mathcal{N}_{\text{CG}}| 4C_{3/2} \quad (126)$$

for all Z, r, θ . Hence

$$\widetilde{\mathcal{M}}_s^2 \leq B_s \widetilde{\mathcal{M}}_s, \quad (127)$$

and Eqs. (117)–(126) give

$$\widetilde{\mathcal{M}}_s \in L^2(d\mu_{\text{cel}}(Z) dr d\theta). \quad (128)$$

Now define the residual two-state modulus on the physical output/scale carrier:

$$\mathcal{R}_{s_1, s_2}(Z, q) = |q|^4 \mathcal{M}_{s_1}(Z, q) \mathcal{M}_{s_2}(Z, q). \quad (129)$$

Under $q = e^{r+i\theta}$,

$$\widetilde{\mathcal{M}}_{s_1}(Z; r, \theta) \widetilde{\mathcal{M}}_{s_2}(Z; r, \theta) = e^{4r} \mathcal{R}_{s_1, s_2}(Z, e^{r+i\theta}). \quad (130)$$

The left side is L^1 by Cauchy–Schwarz and Eq. (128). The fiber log-scale measure is

$$d\mu_{\log}^{\text{fib}}(r, \theta) = e^{4r} dr d\theta.$$

Indeed, for $q = e^{r+i\theta}$,

$$d\mu_{\text{scale}}(q) = |q|^2 d^2 q = e^{2r} (e^{2r} dr d\theta) = e^{4r} dr d\theta = d\mu_{\log}^{\text{fib}}(r, \theta). \quad (131)$$

Thus the map $(r, \theta) \mapsto e^{r+i\theta}$ is measure preserving from $d\mu_{\log}^{\text{fib}}$ to $d\mu_{\text{scale}}$. Therefore Eq. (130) gives

$$\mathcal{R}_{s_1, s_2} \in L^1(d\mu_{\text{cel}}(Z) d\mu_{\text{scale}}(q)). \quad (132)$$

We can now prove integrability of the split four-point kernel itself. On split coordinates

$$((Z, q), (u, v)) \in (\mathbb{C}_Z \times \mathbb{C}_q^\times) \times (\mathbb{C}_u \times \mathbb{C}_v),$$

with measure

$$d\mu_{\text{split}} = d\mu_{\text{cel}}(Z) d\mu_{\text{scale}}(q) d\mu_{\text{sh}}(u) d\mu_{\text{sh}}(v), \quad (133)$$

define

$$\mathcal{K}_{\text{split}}^{a,b}[s_1, s_2](Z, q; u, v) = |q|^4 \overline{\mathcal{R}_{s_1}^{a,b}(Z; q, u)} \mathcal{R}_{s_2}^{a,b}(Z; q, v). \quad (134)$$

For almost every (Z, q) , Fubini gives

$$\begin{aligned} \int_{\mathbb{C}_u \times \mathbb{C}_v} \left| \mathcal{K}_{\text{split}}^{a,b}[s_1, s_2](Z, q; u, v) \right| d\mu_{\text{sh}}(u) d\mu_{\text{sh}}(v) \\ = |q|^4 \mathcal{M}_{s_1}(Z, q) \mathcal{M}_{s_2}(Z, q) = \mathcal{R}_{s_1, s_2}(Z, q). \end{aligned} \quad (135)$$

Integrating Eq. (135) and using Eq. (132) yields

$$\int \left| \mathcal{K}_{\text{split}}^{a,b}[s_1, s_2] \right| d\mu_{\text{split}} < \infty. \quad (136)$$

It remains to transport this already-integrable physical kernel to the coherent product coordinates. Define the single incidence-fiber product measure

$$d\mu_{\text{inc}}(q, u) := d\mu_{\text{scale}}(q) d\mu_{\text{sh}}(u), \quad (137)$$

and the output-fiber shared measure

$$d\mu_{\text{out, fib}} := d\mu_{\text{cel}}(Z) d\mu_{\text{inc}}(q, u) d\mu_{\text{sh}}(v). \quad (138)$$

Write split coordinates as $((Z, q), (u, v))$. The corresponding output-fiber shared coordinate is obtained by reassociation of the two incidence-fiber entries:

$$\Xi((Z, q), (u, v)) = ((Z, (q, u)), v). \quad (139)$$

Here (q, u) denotes the two-component incidence-fiber coordinate. The map is the composition of product reassociation with the measure-preserving scale–shape coordinate equivalence, and hence

$$\Xi_* \mu_{\text{split}} = \mu_{\text{out, fib}}. \quad (140)$$

On the full-measure regular carrier,

$$\mathcal{K}_{\text{split}}^{a,b}[s_1, s_2] = \mathcal{K}_{\text{red}}^{a,b}[s_1, s_2] \circ \Xi, \quad (141)$$

so Eqs. (136) and (140) imply

$$\int \left| \mathcal{K}_{\text{red}}^{a,b}[s_1, s_2] \right| d\mu_{\text{out, fib}} < \infty. \quad (142)$$

For an output-fiber shared coordinate y , let

$$(\lambda(y), v(y))$$

be its centered shared-incidence scale and right cross-ratio variables. The positive transition-envelope density is explicitly

$$\rho_{\text{tr}}(y) = |\mathcal{N}_{\text{CG}}|^2 |\lambda(y)|^2 |v(y)|^2 |1 + v(y)|^2. \quad (143)$$

On the regular set,

$$\mathcal{K}_{\text{red}}^{a,b}[s_1, s_2] = \rho_{\text{tr}} \mathcal{P}_{a,b}[s_1, s_2]. \quad (144)$$

Therefore

$$\int |\mathcal{P}_{a,b}[s_1, s_2]| d(\rho_{\text{tr}} \mu_{\text{out}, \text{fib}}) = \int |\mathcal{K}_{\text{red}}^{a,b}[s_1, s_2]| d\mu_{\text{out}, \text{fib}} < \infty. \quad (145)$$

Finally, the transition-envelope measure identity is pointwise visible from the densities. By Eq. (56),

$$\lambda(q, u) = -\frac{u(1+u)}{q}, \quad |\lambda(q, u)|^2 |q|^2 = |u|^2 |1+u|^2.$$

Hence, on the regular carrier,

$$\begin{aligned} & \rho_{\text{tr}} d\mu_{\text{scale}}(q) d\mu_{\text{sh}}(u) d\mu_{\text{sh}}(v) \\ &= |\mathcal{N}_{\text{CG}}|^2 |\lambda|^2 |v|^2 |1+v|^2 \frac{|q|^2 d^2 q d^2 u d^2 v}{|u|^4 |1+u|^4 |v|^4 |1+v|^4} \\ &= |\mathcal{N}_{\text{CG}}|^2 d^2 q \frac{d^2 u}{|u|^2 |1+u|^2} \frac{d^2 v}{|v|^2 |1+v|^2} \\ &= |\mathcal{N}_{\text{CG}}|^2 d^2 q d\mu_{\text{cr}}(u) d\mu_{\text{cr}}(v). \end{aligned} \quad (146)$$

Therefore the exact split transition map sends the weighted output-fiber measure to the normalized product carrier:

$$(\Xi_{\text{cr}})_*(\rho_{\text{tr}} \mu_{\text{out}, \text{fib}}) = |\mathcal{N}_{\text{CG}}|^2 d\mu_{\text{cel}}(Z_{\text{out}}) d^2 q d\mu_{\text{cr}}(u) d\mu_{\text{cr}}(v). \quad (147)$$

Explicitly,

$$\Xi_{\text{cr}}((Z, (q, u)), v) = ((Z, q), (u, v)), \quad (148)$$

so Ξ_{cr} is the inverse reassociation of Eq. (139) on the regular measured carrier. Since $0 < |\mathcal{N}_{\text{CG}}|^2 < \infty$, Eq. (103) follows from Eq. (145).

Fubini and the independent u, v integrations are therefore justified and give

$$\begin{aligned} \mathfrak{Q}_{\text{coh}}^{a,b}(s_1, s_2) &= |\mathcal{N}_{\text{CG}}|^2 \int_{\mathbb{C}_{Z_{\text{out}}} \times \mathbb{C}_q^\times} \overline{\Phi_{s_1}^{a,b}(Z_{\text{out}}, q)} \Phi_{s_2}^{a,b}(Z_{\text{out}}, q) \\ &\quad \times d\mu_{\text{cel}}(Z_{\text{out}}) d^2 q. \end{aligned} \quad (149)$$

Taking $s_1 = s_2 = s$ yields

$$|\mathcal{N}_{\text{CG}}|^2 \int |\Phi_s^{a,b}(Z_{\text{out}}, q)|^2 d\mu_{\text{cel}}(Z_{\text{out}}) d^2 q < \infty, \quad (150)$$

so

$$\boxed{\Phi_s^{a,b} \in \mathcal{H}_{\text{shape}} \quad \forall s \in \mathcal{D}_{\text{pair}}.} \quad (151)$$

The normalized coherent analysis state is therefore

$$\mathcal{C}_{\text{sh}}^{a,b} s := [\mathcal{N}_{\text{CG}} \Phi_s^{a,b}]_{L^2} \in \mathcal{H}_{\text{shape}}, \quad (152)$$

and Eqs. (100)–(149) give

$$\left\langle \mathcal{C}_{\text{sh}}^{a,b} s_1, \mathcal{C}_{\text{sh}}^{a,b} s_2 \right\rangle_{\mathcal{H}_{\text{shape}}} = \mathfrak{Q}_{\text{coh}}^{a,b}(s_1, s_2). \quad (153)$$

11.3.2 The independently constructed global-flat analysis

Let

$$q(r, \theta) = e^{r+i\theta}, \quad (r, \theta) \in \mathbb{R} \times (-\pi, \pi). \quad (154)$$

The global-flat state is constructed independently of the coherent amplitude.

For fixed output Z_{out} , let

$$\mathfrak{J}_{Z_{\text{out}}}(q, u) = \left(Z_{\text{out}} + \frac{q}{1+u}, Z_{\text{out}} + \frac{q}{u} \right)$$

be the physical incidence chart. Let $\mathcal{K}_{\text{inc}}^{a,b}(Z_{\text{out}}; q, u)$ denote the fixed regulator-free physical incidence kernel. The source-side regulator-free incidence function is

$$\mathcal{R}_s^{a,b}(Z_{\text{out}}; q, u) = \mathbf{1}_{\Omega_{\text{inc}}}(q, u) \Psi_s^{a,b}(\mathfrak{J}_{Z_{\text{out}}}(q, u)) \mathcal{K}_{\text{inc}}^{a,b}(Z_{\text{out}}; q, u), \quad (155)$$

where Ω_{inc} is the regular incidence source set.

The collapsed scale amplitude is defined from this raw incidence function by

$$\mathcal{A}_{\text{coll}}^{a,b}[s](Z_{\text{out}}, q) = \int_{\mathbb{C}} \mathcal{R}_s^{a,b}(Z_{\text{out}}; q, u) d\mu_{\text{sh}}(u). \quad (156)$$

This definition precedes any identification with $\Phi_s^{a,b}$.

The flat logarithmic input is then

$$\mathcal{U}_{\text{flat}}^{a,b}[s](Z_{\text{out}}; r, \theta) := e^{4r} \mathcal{A}_{\text{coll}}^{a,b}[s](Z_{\text{out}}, e^{r+i\theta}). \quad (157)$$

The factor e^{4r} is the exact log-scale density.

Only after this independent construction is in place do we use the coherent identification. The incidence reduction gives, for $q \neq 0$,

$$\mathcal{A}_{\text{coll}}^{a,b}[s](Z_{\text{out}}, q) = |\mathcal{N}_{\text{CG}}| |q|^{-3} \Phi_s^{a,b}(Z_{\text{out}}, q). \quad (158)$$

Substituting $q = e^{r+i\theta}$ into Eqs. (157) and (158) gives the separate identification theorem

$$\mathcal{U}_{\text{flat}}^{a,b}[s](Z_{\text{out}}; r, \theta) = |\mathcal{N}_{\text{CG}}| e^r \Phi_s^{a,b}(Z_{\text{out}}, e^{r+i\theta}). \quad (159)$$

Its completed carrier is

$$\mathcal{K}_{\text{flat}} = L^2(\mathbb{C}_{Z_{\text{out}}} \times \mathbb{R}_r \times (-\pi, \pi)_\theta, d\mu_{\text{cel}}(Z_{\text{out}}) dr d\theta). \quad (160)$$

The regulator-free global incidence input belongs to this space for every core state. Define

$$\mathcal{F}_{\text{flat}}^{a,b} : \mathcal{D}_{\text{pair}} \longrightarrow \mathcal{K}_{\text{flat}}, \quad \mathcal{F}_{\text{flat}}^{a,b} s = [\mathcal{U}_{\text{flat}}^{a,b}[s]]_{L^2}. \quad (161)$$

Its literal polarized pairing is

$$\begin{aligned} \mathfrak{Q}_{\text{flat}}^{a,b}(s_1, s_2) &= \int_{\mathbb{C}} \int_{\mathbb{R}} \int_{-\pi}^{\pi} \overline{\mathcal{U}_{\text{flat}}^{a,b}[s_1](Z; r, \theta)} \mathcal{U}_{\text{flat}}^{a,b}[s_2](Z; r, \theta) \\ &\quad \times d\theta dr d\mu_{\text{cel}}(Z), \end{aligned} \quad (162)$$

and by the L^2 representative theorem

$$\left\langle \mathcal{F}_{\text{flat}}^{a,b} s_1, \mathcal{F}_{\text{flat}}^{a,b} s_2 \right\rangle_{\mathcal{K}_{\text{flat}}} = \mathfrak{Q}_{\text{flat}}^{a,b}(s_1, s_2). \quad (163)$$

The exact source-side transition calculation identifies the coherent and global-flat polarized forms. Equivalently, using Eq. (159) and

$$d^2 q = e^{2r} dr d\theta,$$

one obtains

$$\mathfrak{Q}_{\text{coh}}^{a,b}(s_1, s_2) = \mathfrak{Q}_{\text{flat}}^{a,b}(s_1, s_2). \quad (164)$$

The equality in Eq. (164) is obtained through the same source-space Naimark pairing on the reduced incidence carrier; Eq. (164) is its Hilbert-space consequence.

Combining Eqs. (153), (163), and (164) gives the exact Gram identity

$$\left\langle \mathcal{C}_{\text{sh}}^{a,b} s_1, \mathcal{C}_{\text{sh}}^{a,b} s_2 \right\rangle = \left\langle \mathcal{F}_{\text{flat}}^{a,b} s_1, \mathcal{F}_{\text{flat}}^{a,b} s_2 \right\rangle. \quad (165)$$

11.3.3 Source-kernel inclusion and quotient descent

The global-flat analysis factors through the fixed-external-label physical source by a direct source-space zero-transport argument.

If

$$\iota_{a,b} s = 0,$$

then the source-space representative is the zero L^2 representative:

$$\Psi_s^{a,b} = 0 \quad \text{for } d^2 z_1 d^2 z_2\text{-a.e. } (z_1, z_2). \quad (166)$$

For each fixed output Z_{out} , the incidence chart $\mathfrak{I}_{Z_{\text{out}}}$ is measure preserving from the physical incidence carrier to the ordered source-pair carrier. Pulling Eq. (166) through this chart and multiplying by the fixed incidence kernel therefore gives

$$\mathcal{R}_s^{a,b}(Z_{\text{out}}; q, u) = 0 \quad \text{for incidence-a.e. } (q, u). \quad (167)$$

The exact scale/shape coordinate equivalence transports Eq. (167) to the product scale–shape measure. Fubini then gives

$$\mathcal{A}_{\text{coll}}^{a,b}[s](Z_{\text{out}}, q) = 0 \quad \text{for } d\mu_{\text{scale}}(q)\text{-a.e. } q. \quad (168)$$

The logarithmic map

$$(r, \theta) \mapsto e^{r+i\theta}$$

is measure preserving from the fiber log-scale measure to the physical scale measure, and the flat log-cylinder measure is absolutely continuous with respect to that fiber measure. Therefore

$$\mathcal{A}_{\text{coll}}^{a,b}[s](Z_{\text{out}}, e^{r+i\theta}) = 0 \quad \text{for flat-log-a.e. } (r, \theta). \quad (169)$$

Multiplication by the positive log-scale density in Eq. (157) preserves this zero set, so

$$\mathcal{U}_{\text{flat}}^{a,b}[s](Z_{\text{out}}; r, \theta) = 0 \quad \text{for flat-log-a.e. } (r, \theta). \quad (170)$$

Fubini in the output variable yields zero almost everywhere on the full product carrier. Passing to the L^2 class gives

$$\iota_{a,b}s = 0 \implies \mathcal{F}_{\text{flat}}^{a,b}s = 0. \quad (171)$$

Therefore, for $s \in \ker \iota_{a,b}$,

$$\begin{aligned} \|\mathcal{C}_{\text{sh}}^{a,b}s\|^2 &= \|\mathcal{F}_{\text{flat}}^{a,b}s\|^2 && \text{by Eq. (165)} \\ &= 0 && \text{by Eq. (171)}. \end{aligned} \quad (172)$$

Hence

$$\ker \iota_{a,b} \subseteq \ker \mathcal{C}_{\text{sh}}^{a,b}. \quad (173)$$

If

$$\iota_{a,b}s_1 = \iota_{a,b}s_2,$$

then $s_1 - s_2 \in \ker \iota_{a,b}$, so Eq. (173) implies

$$\mathcal{C}_{\text{sh}}^{a,b}s_1 = \mathcal{C}_{\text{sh}}^{a,b}s_2.$$

Since $\mathcal{N}_{\text{CG}} \neq 0$,

$$\Phi_{s_1}^{a,b} = \Phi_{s_2}^{a,b} \quad \text{for } d\mu_{\text{cel}}(Z_{\text{out}}) d^2q\text{-a.e. } (Z_{\text{out}}, q). \quad (174)$$

The fixed-external-label source map has the canonical quotient/range equivalence

$$\mathcal{D}_{\text{pair}} / \ker \iota_{a,b} \simeq \mathcal{R}_{\text{phys}}^{a,b}. \quad (175)$$

The kernel inclusion in Eq. (173) therefore gives a unique descended linear map

$$\boxed{T_{\text{sh}}^{a,b} : \mathcal{R}_{\text{phys}}^{a,b} \longrightarrow \mathcal{H}_{\text{shape}}} \quad (176)$$

with

$$T_{\text{sh}}^{a,b}(\iota_{a,b}s) = [\mathcal{N}_{\text{CG}}\Phi_s^{a,b}]_{L^2}. \quad (177)$$

Thus $T_{\text{sh}}^{a,b}$ is a function of physical source equality and not of a chosen ordered-core presentation. The full-signed and retained presentation maps satisfy the analogous source-kernel inclusions.

11.4 Uniform quadratic-Gaussian spatial mass

For $\epsilon > 0$, define

$$B_{a,\epsilon} = C_{a,a} \int_{\mathbb{C}} K_{\epsilon}(\xi) |\xi|^{-1} d^2\xi, \quad (178)$$

where $C_{a,a}$ is the uniform bound for the equal-label diagonal homogeneous kernel. The integral in Eq. (178) is finite.

The equal-label Gaussian spectral identity and

$$|J_{a,a}(\xi t)| \leq C_{a,a}$$

give, for every $t \neq 0$ and every anchor $A \in \mathbb{C}$,

$$|t| \int_{\mathbb{C}} e^{-\epsilon|t(s^2+s)-A|^2} d^2s \leq B_{a,\epsilon}. \quad (179)$$

The critical point of $s \mapsto s^2 + s$ therefore creates no Fubini singularity.

For every X in the Haar L^1 domain,

$$\begin{aligned} \int_{\mathbb{C}^\times} \int_{\mathbb{C}} |H(X, t)| |t| e^{-\epsilon|t(s^2+s)-A|^2} d^2s d\mu_\times(t) \\ \leq B_{a,\epsilon} \int_{\mathbb{C}^\times} |H(X, t)| d\mu_\times(t) < \infty. \end{aligned} \quad (180)$$

If $A_{m,w}$ is the harmonic anchor observer used below, then

$$\begin{aligned} \int_{\mathbb{C}} |A_{m,w}(A)| \int_{\mathbb{C}^\times} \int_{\mathbb{C}} |H(X, t)| |t| e^{-2w|t(s^2+s)-A|^2} d^2s d\mu_\times(t) d^2A \\ \leq B_{a,2w} \|A_{m,w}\|_{L^1(\mathbb{C})} \|H(X, \cdot)\|_{L^1(d\mu_\times)} < \infty. \end{aligned} \quad (181)$$

Equations (180)–(181) are the absolute Fubini estimates used in the $J \rightarrow \mathcal{Q} \rightarrow \mathcal{G}$ portion of the proof.

12 Data and conventions for the uniqueness theorem

This section fixes every convention used in the source-separation argument.

12.1 Naimark character and external-label differences

For $z \neq 0$, $\alpha \in \mathbb{C}$ and $m \in \mathbb{Z}$, define

$$\mathbf{N}(z; \alpha, m) = |z|^\alpha \left(\frac{z}{|z|} \right)^m. \quad (182)$$

The radial power uses the positive real radius $|z|$, and the phase exponent is the ordinary integer power. Values assigned at $z = 0$ are irrelevant to the measured identities below because the puncture is null for the measures in use.

For external labels $a = (j_a, f_a)$ and $b = (j_b, f_b)$, use the principal-series Mellin frequencies

$$\nu_a = 2\pi f_a, \quad \nu_b = 2\pi f_b, \quad (183)$$

and define

$$\delta\nu_{ab} = \nu_a - \nu_b, \quad \delta j_{ab} = j_a - j_b. \quad (184)$$

The unit input-chord character is

$$\chi_{a,b}^{\text{in}}(t) = \begin{cases} 1, & t = 0, \\ \mathbf{N}(t; i\delta\nu_{ab}, \delta j_{ab}), & t \neq 0. \end{cases} \quad (185)$$

It satisfies $|\chi_{a,b}^{\text{in}}(t)| = 1$.

12.2 Output center, scale and shape carrier

We reserve

$$Z_{\text{out}}$$

for the celestial output center throughout the uniqueness proof. The north-chart round-sphere measure is

$$d\mu_{\text{cel}}(Z_{\text{out}}) = \frac{4 d^2 Z_{\text{out}}}{(1 + |Z_{\text{out}}|^2)^2}. \quad (186)$$

The output scale q carries flat complex Lebesgue measure on $\mathbb{C}^\times$; omitting $q = 0$ changes no L^2 class. The one-shape output Hilbert space is Eq. (59):

$$\mathcal{H}_{\text{shape}} = L^2\left(\mathbb{C}_{Z_{\text{out}}} \times \mathbb{C}_q^\times, d\mu_{\text{cel}}(Z_{\text{out}}) d^2 q\right).$$

For a core source s , let $\Phi_s^{a,b}(Z_{\text{out}}, q)$ be the coherent shape amplitude of Eq. (58). The intrinsic shape transform is represented by

$$T_{\text{sh}}^{a,b}(\iota_{a,b}s) = \left[\mathcal{N}_{\text{CG}} \Phi_s^{a,b}\right]_{L^2}, \quad (187)$$

where $\mathcal{N}_{\text{CG}} \neq 0$ is the fixed Clebsch–Gordan normalization. Thus

$$T_{\text{sh}}^{a,b}(\iota_{a,b}s) = 0 \iff \Phi_s^{a,b}(Z_{\text{out}}, q) = 0 \text{ for } d\mu_{\text{cel}}(Z_{\text{out}}) d^2 q\text{-a.e. } (Z_{\text{out}}, q). \quad (188)$$

This is the functional meaning of “physical output nullity” used below.

12.3 Fixed-output incidence and cross-ratio measure

The physical cross-ratio measure is

$$d\mu_{\text{cr}}(u) = \mathbf{1}_{\mathbb{C} \setminus \{0, -1\}}(u) \frac{d^2 u}{|u|^2 |1 + u|^2}. \quad (189)$$

At fixed Z_{out} , the incidence map is

$$\mathfrak{I}_{Z_{\text{out}}}(q, u) = \left(Z_{\text{out}} + \frac{q}{1 + u}, Z_{\text{out}} + \frac{q}{u}\right), \quad (190)$$

with inverse

$$q(Z_{\text{out}}; z_1, z_2) = -\frac{(Z_{\text{out}} - z_1)(Z_{\text{out}} - z_2)}{z_1 - z_2}, \quad u(Z_{\text{out}}; z_1, z_2) = \frac{Z_{\text{out}} - z_1}{z_1 - z_2} \quad (191)$$

on the regular source-pair locus.

The exact coarea identity is

$$(\mathfrak{I}_{Z_{\text{out}}})_* \left(d^2 q \frac{d^2 u}{|u|^2 |1 + u|^2} \right) = \frac{d^2 z_1 d^2 z_2}{|z_1 - z_2|^2}. \quad (192)$$

12.4 Original source, dephased source and coarea phase

Let

$$\Psi_s^{a,b}(z_1, z_2) \quad (193)$$

denote the source-coordinate ordered-pair representative of the fixed-external-label source. The normalized baseline phase factorizes on the regular incidence locus as

$$\eta_{a,b}(\lambda, u) = \eta_{a,b}^{\text{sc}}(\lambda) \eta_{a,b}^{\text{cr}}(u), \quad (194)$$

where the fixed positive normalization has unit phase.

Since

$$z_1 - z_2 = -\frac{q}{u(1+u)}, \quad \lambda(q, u) = -\frac{u(1+u)}{q} = (z_1 - z_2)^{-1}, \quad (195)$$

define the dephased ordered-pair source by

$$\tilde{\Psi}_s^{a,b}(z_1, z_2) = \eta_{a,b}^{\text{sc}}((z_1 - z_2)^{-1}) \Psi_s^{a,b}(z_1, z_2). \quad (196)$$

Dephasing changes only a unit phase off the pair diagonal.

The remaining cross-ratio phase is

$$\chi_{a,b}(Z_{\text{out}}; z_1, z_2) = \mathbf{N}\left(-\frac{z_2 - Z_{\text{out}}}{z_1 - Z_{\text{out}}}; i\delta\nu_{ab}, \delta j_{ab}\right). \quad (197)$$

Indeed,

$$-\frac{z_2 - Z_{\text{out}}}{z_1 - Z_{\text{out}}} = -\frac{1+u}{u},$$

and the phase-twisted incidence source obeys the exact identity

$$\eta_{a,b}(\lambda(q, u), u) \Psi_s^{a,b}(z_1, z_2) = \chi_{a,b}(Z_{\text{out}}; z_1, z_2) \tilde{\Psi}_s^{a,b}(z_1, z_2). \quad (198)$$

Equation (198) is the identity used when the shape integral is transported by coarea.

12.5 Source windows

For $n \in \mathbb{N}$, define

$$W_n = \left\{ (z_1, z_2) : \|(z_1, z_2)\| \leq n, \quad |z_1 - z_2| \geq \frac{1}{n+1} \right\}, \quad (199)$$

and

$$S_{s,n}^{a,b}(z_1, z_2) = \mathbf{1}_{W_n}(z_1, z_2) \tilde{\Psi}_s^{a,b}(z_1, z_2). \quad (200)$$

These windows exhaust every off-diagonal ordered source pair.

The corresponding windowed coherent output is

$$\Phi_{s,n}^{a,b}(Z_{\text{out}}, q) = \int_{\mathbb{C}} \mathbf{1}_{W_n}(\mathfrak{I}_{Z_{\text{out}}}(q, u)) \mathcal{I}_{a,b,s}(Z_{\text{out}}, q; u) d\mu_{\text{cr}}(u). \quad (201)$$

For almost every output point,

$$\Phi_{s,n}^{a,b}(Z_{\text{out}}, q) \longrightarrow \Phi_s^{a,b}(Z_{\text{out}}, q).$$

Moreover $\mathbf{1}_{W_n} \rightarrow 1$ almost everywhere off the null diagonal and

$$|\mathbf{1}_{W_n}(\mathfrak{I}_{Z_{\text{out}}}(q, u)) \mathcal{J}_{a,b,s}(Z_{\text{out}}, q; u)| \leq |\mathcal{J}_{a,b,s}(Z_{\text{out}}, q; u)|.$$

The doubled-coherent L^1 majorant established in Section 11.3 therefore gives dominated convergence for every admissible Gaussian/Schwartz observer introduced below, so the corresponding bilinear observer pairings converge to those of $\Phi_s^{a,b}$.

12.6 Source base point, chord, Haar and logarithmic-cylinder sources

Using the center/chord coordinates of Eq. (87) and the field $G_{a,b,s}$ of Eq. (89), the multiplicative-Haar normalization is

$$H_{a,b,s}(X, t) = |t|^2 G_{a,b,s}(X, t). \quad (202)$$

With

$$d\mu_{\times}(t) = \frac{d^2 t}{|t|^2}, \quad (203)$$

we have $H d\mu_{\times} = G d^2 t$ off the null puncture $t = 0$.

Under $t = e^{r+i\theta}$,

$$d\mu_{\times}(t) = dr d\theta, \quad (204)$$

and the weighted Mellin-cylinder source is

$$F_X(r, \theta) = e^{-r/2} H_{a,b,s}(X, e^{r+i\theta}) = e^{3r/2} G_{a,b,s}(X, e^{r+i\theta}). \quad (205)$$

12.7 Damped quadratic pairing, Schwinger value and diagonal kernel

For $\delta > 0$, anchor $A_0 \in \mathbb{C}$, and quadratic/linear coefficients $Q, L \in \mathbb{C}$, define the absolutely convergent pairing

$$\mathcal{W}_{\delta}^{a,b}(A_0; Q, L) = \int_{\mathbb{C}} (\chi_{a,b}^{\text{in}}(s))^2 \exp(-\delta|A_0 - s|^2 + i \operatorname{Re}(Qs^2 + Ls)) d^2 s. \quad (206)$$

The homogeneous value used below is defined by an absolutely convergent Schwinger integral before any zero-damping limit is taken. Set

$$n_{ab} = 2\delta j_{ab}, \quad \mu_{ab} = 2\delta \nu_{ab}, \quad h_{ab} = \frac{|n_{ab}|}{2}, \quad (207)$$

and

$$p_{ab} = \frac{|n_{ab}| + 2 - i\mu_{ab}}{2}, \quad \operatorname{Re} p_{ab} = \frac{|n_{ab}| + 2}{2} > 0. \quad (208)$$

Let

$$\Pi_n(z) = \begin{cases} z^{|n|}, & n \geq 0, \\ \bar{z}^{|n|}, & n < 0, \end{cases} \quad (209)$$

and define the compensated Schwinger slice

$$\mathfrak{S}_{n,\mu}(z, \tau) = \frac{\tau^{(|n|+2-i\mu)/2-1}}{\Gamma((|n|+2-i\mu)/2)} |z|^2 \Pi_n(z) e^{-\tau|z|^2}, \quad \tau > 0. \quad (210)$$

For $z \neq 0$,

$$\int_0^\infty \mathfrak{S}_{n,\mu}(z, \tau) d\tau = |z|^{i\mu} e^{in \arg z}. \quad (211)$$

In particular, Eq. (211) recovers $(\chi_{a,b}^{\text{in}}(z))^2$ for $(n, \mu) = (n_{ab}, \mu_{ab})$.

For $A > 0$ and $B \in \mathbb{C}$, define the evaluated Schwinger moment

$$\mathcal{M}_{a,b}(\tau; A, B) = \int_{\mathbb{C}} \mathfrak{S}_{n_{ab}, \mu_{ab}}(z, \tau) e^{i \operatorname{Re}(Az^2 + Bz)} d^2 z. \quad (212)$$

At every $\tau > 0$, this is an absolutely convergent Gaussian-polynomial integral. Its finite Gaussian-moment evaluation is used only through the estimates displayed below.

For $Q \neq 0$, let

$$r_Q = e^{-i \arg(Q)/2}, \quad |r_Q| = 1, \quad Q r_Q^2 = |Q|. \quad (213)$$

We define

$$\mathcal{H}_{a,b}(Q, L) = (\chi_{a,b}^{\text{in}}(r_Q))^2 \int_0^\infty \mathcal{M}_{a,b}(\tau; |Q|, L r_Q) d\tau. \quad (214)$$

The integral in Eq. (214) is absolutely convergent. To see this, put

$$A = |Q|, \quad B = |L|.$$

The physical modes satisfy $|n_{ab}| = 2h_{ab}$. For $0 < \tau \leq A$,

$$\begin{aligned} |\mathcal{M}_{a,b}(\tau; A, L r_Q)| &\leq C_{a,b}^{(0)} \frac{(1 + B/\sqrt{A})^{2h_{ab}+2}}{A^{h_{ab}+2}} \\ &\quad \times \tau^{h_{ab}} \exp\left(-\frac{B^2}{8A^2}\tau\right), \end{aligned} \quad (215)$$

while for $\tau \geq A$,

$$|\mathcal{M}_{a,b}(\tau; A, L r_Q)| \leq C_{a,b}^{(0)} \frac{1}{\tau^2} \left(1 + \frac{B}{\sqrt{\tau}}\right)^{2h_{ab}+2} \exp\left(-\frac{B^2}{8\tau}\right). \quad (216)$$

Both bounds come from the explicitly evaluated Gaussian moments after the unit rotation in Eq. (213). Their piecewise majorant is integrable on $(0, \infty)$, and its complete integral obeys

$$\int_0^\infty m_{h_{ab}}(\tau; A, B) d\tau \leq \frac{C_{a,b}^{(1)}}{A}, \quad (217)$$

where $C_{a,b}^{(1)}$ is independent of B . Consequently

$$|\mathcal{H}_{a,b}(Q, L)| \leq \frac{C_{a,b}}{|Q|} \quad (Q \neq 0), \quad (218)$$

with $C_{a,b}$ independent of L .

The Schwinger value in Eq. (214) is the zero-damping Abel limit of Eq. (206), independent of the fixed anchor A_0 .

Proposition (Abel–Fresnel convergence). For every $A_0 \in \mathbb{C}$, $Q \neq 0$, and $L \in \mathbb{C}$,

$$\lim_{\delta \downarrow 0} \mathcal{W}_\delta^{a,b}(A_0; Q, L) = \mathcal{H}_{a,b}(Q, L). \quad (219)$$

Moreover the bound in Eq. (218) holds uniformly in L .

Proof. Rotate $s = r_Q z$. Lebesgue measure is invariant, the squared input character contributes the unit factor $(\chi_{a,b}^{\text{in}}(r_Q))^2$, and the quadratic coefficient becomes $|Q|$. The damped anchor becomes $r_Q^{-1} A_0$. Insert Eq. (211) before integrating in z . For every $\delta > 0$, compensated Schwinger Fubini is absolute because the translated Gaussian observer is $L^1(\mathbb{C})$. The z -integral is the same finite Gaussian-polynomial moment as in Eq. (212), with the radial damping shifted by δ and the corresponding finite linear term generated by the fixed anchor.

At each fixed $\tau > 0$, the damped evaluated moment converges to $\mathcal{M}_{a,b}(\tau; |Q|, Lr_Q)$ as $\delta \downarrow 0$. The small- and large-scale bounds (215)–(216) supply an integrable dominating function, uniformly for sufficiently small $\delta > 0$. Dominated convergence in the Schwinger parameter therefore yields Eq. (219). The same majorant integrated over $(0, \infty)$ gives Eq. (218).

The diagonal homogeneous kernel is

$$J_{a,b}(c) = |c| \mathcal{H}_{a,b}(c, c), \quad J_{a,b}(0) = 0. \quad (220)$$

Equation (218) gives

$$|J_{a,b}(c)| \leq C_{a,b} \quad (221)$$

for all $c \in \mathbb{C}$. The kernel is strongly measurable because it is the pointwise Abel limit of measurable damped kernels.

For later Gaussian testing define

$$K_\epsilon(\xi) = \frac{1}{4\pi\epsilon} \exp\left(-\frac{|\xi|^2}{4\epsilon}\right), \quad \epsilon > 0. \quad (222)$$

It has unit mass on $\mathbb{C} \simeq \mathbb{R}^2$. The bound (221) and $K_\epsilon(\xi)|\xi|^{-1} \in L^1(\mathbb{C})$ justify the Gaussian spectral dominated-convergence steps used below.

12.8 Homogeneous kernels and Schwinger parameters

For $k \in \mathbb{Z}$ and $\lambda \in \mathbb{C}$, define

$$K_{k,\lambda}(z) = |z|^{\lambda-2} e^{ik \arg z}, \quad p(k, \lambda) = \frac{|k| + 2 - \lambda}{2}. \quad (223)$$

When $\text{Re } p(k, \lambda) > 0$,

$$K_{k,\lambda}(z) = \frac{|z|^{|k|} e^{ik \arg z}}{\Gamma(p(k, \lambda))} \int_0^\infty a^{p(k, \lambda)-1} e^{-a|z|^2} da. \quad (224)$$

All exchanges of the a -integral with source or spatial integration below are made only after the corresponding absolute-integrability estimate has been established.

13 All-label uniqueness on the intrinsic physical source range

Let $s \in \mathcal{D}_{\text{pair}}$ and suppose its intrinsic full signed principal-series output vanishes. By Eq. (65), this is equivalent to vanishing of the cross-ratio output. We prove that the corresponding source vector $\iota_{a,b}s$ is zero.

13.1 From physical output nullity to the relative-scale kernel

Let $s \in \mathcal{D}_{\text{pair}}$ and suppose

$$T_{\text{full}}^{a,b}(\iota_{a,b}s) = 0.$$

By Eqs. (65) and (188),

$$\Phi_s^{a,b}(Z_{\text{out}}, q) = 0$$

for $d\mu_{\text{cel}}(Z_{\text{out}}) d^2q$ -almost every output point.

13.1.1 Admissible label-dressed output observers

Throughout this source-separation argument we use the bilinear testing functional

$$\mathfrak{B}(\Theta, \Phi) = \int_{\mathbb{C} \times \mathbb{C}} \Theta(Z, q) \Phi(Z, q) d\mu_{\text{cel}}(Z) d^2q. \quad (225)$$

No complex conjugation is inserted in Eq. (225); the Hilbert-space inner product is not being used at this step. This convention fixes the phase of the scale dressing below.

Write

$$\rho_{\text{cel}}(Z) = \frac{4}{(1 + |Z|^2)^2} \quad (226)$$

for the north-chart density. For $\epsilon > 0$, center frequency $\kappa \in \mathbb{C}$, and $\varphi \in \mathcal{S}(\mathbb{C})$, define

$$D_{\epsilon, \kappa}(Z) = \exp\left(-\epsilon|Z|^2 + i \operatorname{Re}(\kappa Z)\right) \quad (227)$$

and

$$\widehat{\varphi}(q) = \int_{\mathbb{C}} e^{i \operatorname{Re}(\xi(\zeta)q)} \varphi(\zeta) d^2\zeta, \quad \xi(\zeta) = -2\pi\bar{\zeta}. \quad (228)$$

The all-label physical observer is

$$\Theta_{\epsilon, \kappa, \varphi}^{a,b}(Z, q) = \rho_{\text{cel}}(Z)^{-1} D_{\epsilon, \kappa}(Z) \chi_{a,b}^{\text{in}}(q) \widehat{\varphi}(q). \quad (229)$$

Because

$$|\chi_{a,b}^{\text{in}}(q)| = 1,$$

the scale-character dressing preserves the L^2 norm of the scale observer. The Gaussian compensates the inverse stereographic weight, so

$$\Theta_{\epsilon, \kappa, \varphi}^{a,b} \in \mathcal{H}_{\text{shape}}.$$

Hence shape nullity gives

$$\mathfrak{B}\left(\Theta_{\epsilon, \kappa, \varphi}^{a,b}, \Phi_s^{a,b}\right) = 0. \quad (230)$$

For a regular incidence point, put

$$t = z_1 - z_2, \quad c = z_2 - Z_{\text{out}}.$$

The coarea phase is

$$\chi_{\text{coarea}} = \chi_{a,b}^{\text{in}} \left(-\frac{c}{c+t} \right), \quad (231)$$

while

$$q = -\frac{c(c+t)}{t}. \quad (232)$$

Multiplicativity of the unit character gives the exact dressing identity

$$\chi_{a,b}^{\text{in}}(q) \chi_{\text{coarea}} = \frac{\chi_{a,b}^{\text{in}}(c)^2}{\chi_{a,b}^{\text{in}}(t)}. \quad (233)$$

Thus the squared center chord character used below is not inserted by hand; it is the product of the original coarea phase with the explicit scale-character factor in Eq. (229).

Using Eq. (228), the coarea formula, and absolute Fubini, the source-window version of Eq. (230) becomes

$$0 = \lim_{n \rightarrow \infty} \int_{\mathbb{C}} \varphi(\zeta) \int_{\mathbb{C}^2} S_{s,n}^{a,b}(z_1, z_2) \frac{\mathcal{F}_{\epsilon; \xi(\zeta), \kappa}^{a,b}(z_1, z_2)}{\chi_{a,b}^{\text{in}}(t)} \frac{d^2 z_1 d^2 z_2}{|t|^2} d^2 \zeta, \quad (234)$$

where

$$\mathcal{F}_{\epsilon; \xi, \kappa}^{a,b}(z_1, z_2) = \int_{\mathbb{C}} D_{\epsilon, \kappa}(Z_{\text{out}}) \chi_{a,b}^{\text{in}}(c)^2 e^{i \text{Re}(\xi q(Z_{\text{out}}; z_1, z_2))} d^2 Z_{\text{out}}. \quad (235)$$

13.1.2 Translated quadratic center integral

Since $Z_{\text{out}} = z_2 - c$,

$$\kappa Z_{\text{out}} + \xi q(Z_{\text{out}}; z_1, z_2) = \kappa z_2 - \frac{\xi}{t} c^2 - (\xi + \kappa) c. \quad (236)$$

Translation invariance gives

$$\mathcal{F}_{\epsilon; \xi, \kappa}^{a,b}(z_1, z_2) = e^{i \text{Re}(\kappa z_2)} \mathcal{W}_{\epsilon}^{a,b} \left(z_2; -\frac{\xi}{t}, -(\xi + \kappa) \right). \quad (237)$$

The anchor z_2 is harmless in the zero-damping limit. Proposition 12.7 therefore gives, for $\xi \neq 0$,

$$\mathcal{F}_{\xi, \kappa}^{a,b}(z_1, z_2) = e^{i \text{Re}(\kappa z_2)} \mathcal{H}_{a,b} \left(-\frac{\xi}{t}, -(\xi + \kappa) \right). \quad (238)$$

For fixed κ , vanishing of the tested pairings for every $\varphi \in \mathcal{S}(\mathbb{C})$, followed by the source-window and zero-damping limits, implies

$$\mathcal{F}_{\xi, \kappa}^{a,b} = 0 \quad \text{in the source-pair coefficient for a.e. } \xi \in \mathbb{C}^{\times}. \quad (239)$$

Only this almost-everywhere statement is used below.

13.1.3 Dilation to the diagonal kernel

On the regular set

$$\xi \neq 0, \quad \xi + \kappa \neq 0,$$

define

$$\sigma = -\frac{(\xi + \kappa)^2}{\xi}. \quad (240)$$

Homogeneous dilation gives

$$\frac{\mathcal{F}_{\xi, \kappa}^{a, b}(z_1, z_2)}{\chi_{a, b}^{\text{in}}(t)} = \frac{\chi_{a, b}^{\text{in}}\left(\frac{\xi + \kappa}{\xi}\right)^2}{|\xi|} \chi_{a, b}^{\text{in}}(t) |t| e^{i \operatorname{Re}(\kappa z_2)} J_{a, b}(\sigma t). \quad (241)$$

The scalar prefactor is nonzero on this regular set.

For fixed κ , the map

$$R_\kappa(\xi) = -\frac{(\xi + \kappa)^2}{\xi} \quad (242)$$

is a finite-to-one holomorphic map away from $\xi = 0$ and its finite critical set. On each regular branch it is a smooth local diffeomorphism with nonzero real Jacobian. The area formula therefore transports conull sets between the ξ - and σ -planes, up to the finite set of critical values. Consequently Eq. (239) yields the joint measurable statement

$$\boxed{\mathcal{C}_{a, b, s}(\kappa, \sigma) = 0 \quad \text{for a.e. } (\kappa, \sigma) \in \mathbb{C} \times \mathbb{C},} \quad (243)$$

where

$$\mathcal{C}_{a, b, s}(\kappa, \sigma) = \int_{\mathbb{C}^2} \tilde{\Psi}_s^{a, b}(z_1, z_2) \frac{\chi_{a, b}^{\text{in}}(t)}{|t|} e^{i \operatorname{Re}(\kappa z_2)} J_{a, b}(\sigma t) d^2 z_1 d^2 z_2. \quad (244)$$

The null sets $t = 0$, $\xi = 0$, $\xi + \kappa = 0$, and the critical values of R_κ do not affect Eq. (243).

13.1.4 Source base point and center Fourier separation

Using the center/chord coordinates of Eq. (87) and Eq. (89),

$$\mathcal{C}_{a, b, s}(\kappa, \sigma) = \int_{\mathbb{C}} e^{i \operatorname{Re}(\kappa X)} \left[\int_{\mathbb{C}} G_{a, b, s}(X, t) J_{a, b}(\sigma t) d^2 t \right] d^2 X. \quad (245)$$

Joint measurability and Eq. (243) imply that for almost every σ , this Fourier transform vanishes for almost every κ . For fixed such σ , the bracket in Eq. (245) belongs to $L^1(\mathbb{C}_X)$; its Fourier transform is therefore continuous in κ . Vanishing almost everywhere in κ upgrades to vanishing everywhere, and complex L^1 Fourier uniqueness gives one conull set $E_0 \subset \mathbb{C}$ such that

$$\int_{\mathbb{C}} G_{a, b, s}(X, t) J_{a, b}(\sigma t) d^2 t = 0 \quad (246)$$

for every $X \in E_0$ and almost every $\sigma \in \mathbb{C}$.

The passage to the Haar-normalized source is

$$H(X, t) = |t|^2 G(X, t), \quad H d\mu_\times = G d^2 t, \quad (247)$$

and for almost every X ,

$$\int_{\mathbb{C}^\times} |t|^{-1/2} |H(X, t)| d\mu_\times(t) < \infty. \quad (248)$$

13.2 From $J_{a,b}$ -nullity to translated quadratic Gaussian nullity

The diagonal kernel has the exact Gaussian spectral identity

$$\int_{\mathbb{C}} K_{\epsilon}(\xi) e^{-i \operatorname{Re}(\xi A)} \frac{J_{a,b}(\xi)}{|\xi|} d^2 \xi = \int_{\mathbb{C}} (\chi_{a,b}^{\text{in}}(s))^2 e^{-\epsilon |s^2 + s - A|^2} d^2 s, \quad (249)$$

for every $\epsilon > 0$ and $A \in \mathbb{C}$. The spectral side is absolutely integrable because

$$K_{\epsilon}(\xi) |\xi|^{-1} \in L^1(\mathbb{C})$$

and $J_{a,b}$ is bounded.

For $t \neq 0$, dilation gives

$$\begin{aligned} & \int_{\mathbb{C}} K_{\epsilon}(\xi) e^{-i \operatorname{Re}(\xi A)} \frac{J_{a,b}(\xi t)}{|\xi|} d^2 \xi \\ &= |t| \int_{\mathbb{C}} (\chi_{a,b}^{\text{in}}(s))^2 e^{-\epsilon |t(s^2 + s) - A|^2} d^2 s. \end{aligned} \quad (250)$$

The absolute source/spectral bound permits Fubini with $H(X, t) d\mu_{\times}(t)$. Since Eq. (246) holds for almost every spectral frequency, the spectral side vanishes after integration in t . Thus, on a common conull set $E_1 \subset E_0$,

$$\mathcal{Q}_{X,\epsilon}(A) := \int_{\mathbb{C} \times \mathbb{C}} H(X, t) |t| (\chi_{a,b}^{\text{in}}(s))^2 e^{-\epsilon |t(s^2 + s) - A|^2} d^2 s d\mu_{\times}(t) = 0 \quad (251)$$

for every $\epsilon > 0$ and every anchor $A \in \mathbb{C}$.

13.3 Harmonic Gaussian testing

For $m \in \mathbb{Z}$, define

$$P_m(z) = \begin{cases} z^{|m|}, & m \geq 0, \\ \bar{z}^{|m|}, & m < 0. \end{cases} \quad (252)$$

For $w > 0$, let

$$A_{m,w}(A) = P_m(A) e^{-2w|A|^2}. \quad (253)$$

Completing the square gives

$$2w|A|^2 + 2w|y - A|^2 = 4w \left| A - \frac{y}{2} \right|^2 + w|y|^2, \quad (254)$$

and rotational symmetry yields

$$\int_{\mathbb{C}} A_{m,w}(A) e^{-2w|y - A|^2} d^2 A = c_{m,w} P_m(y) e^{-w|y|^2}, \quad c_{m,w} = \frac{\pi}{4w} 2^{-|m|} \neq 0. \quad (255)$$

Define the harmonic Gaussian source moment

$$\begin{aligned} \mathcal{G}_{X,m,w} &:= \int_{\mathbb{C} \times \mathbb{C}} H(X, t) |t| (\chi_{a,b}^{\text{in}}(s))^2 \\ &\quad \times P_m(t(s^2 + s)) e^{-w|t(s^2 + s)|^2} d^2 s d\mu_{\times}(t). \end{aligned} \quad (256)$$

The source/anchor/spatial field is absolutely integrable. Indeed, Eq. (179) gives a uniform bound on the spatial Gaussian after multiplication by $|t|$, and Eq. (181) gives the full source/anchor/spatial L^1 estimate required for three-variable Fubini.

For $X \in E_1$, multiply

$$\mathcal{Q}_{X,2w}(A) = 0$$

by $A_{m,w}(A)$ and integrate in A . Absolute three-variable Fubini and Eq. (255) give

$$\begin{aligned} 0 &= \int_{\mathbb{C}} A_{m,w}(A) \mathcal{Q}_{X,2w}(A) d^2 A \\ &= c_{m,w} \mathcal{G}_{X,m,w}. \end{aligned} \quad (257)$$

Since $c_{m,w} \neq 0$,

$$X \in E_1 \implies \mathcal{G}_{X,m,w} = 0 \quad \forall m \in \mathbb{Z}, \forall w > 0. \quad (258)$$

The conull source-base set is selected before the Mellin frequency is introduced, so the subsequent all- ω statement does not require an uncountable intersection of exceptional sets.

13.4 Shifted Mellin factorization

For $X \in E_1$, define

$$M_{a,b,s}(X; m, \omega) = \int_{\mathbb{C}^\times} H_{a,b,s}(X, t) |t| \mathbf{N}\left(t; -\frac{3}{2} + i\omega, m\right) d\mu_\times(t). \quad (259)$$

By Eq. (182),

$$\left| |t| \mathbf{N}\left(t; -\frac{3}{2} + i\omega, m\right) \right| = |t|^{-1/2}, \quad (260)$$

so Eq. (248) gives absolute convergence for every m, ω .

The remaining physical shape variable enters through $s^2 + s = s(1 + s)$. Define

$$B_{a,b}^{m,\omega}(s) = (\chi_{a,b}^{\text{in}}(s))^2 \mathbf{N}\left(s^2 + s; -\frac{3}{2} + i\omega, m\right) \quad (261)$$

and

$$\beta_{a,b}(m, \omega) = \int_{\mathbb{C}} B_{a,b}^{m,\omega}(s) d^2 s. \quad (262)$$

On the punctured plane,

$$|B_{a,b}^{m,\omega}(s)| = |s|^{-3/2} |1 + s|^{-3/2}, \quad (263)$$

which is integrable near 0, near -1 , and at infinity.

For the joint field

$$\begin{aligned} \mathcal{F}_{X,m,\omega}(t, s) &:= H_{a,b,s}(X, t) |t| (\chi_{a,b}^{\text{in}}(s))^2 \\ &\quad \times \mathbf{N}\left(t(s^2 + s); -\frac{3}{2} + i\omega, m\right), \end{aligned} \quad (264)$$

multiplicativity of $\mathbf{N}$ gives the pointwise factorization

$$\mathcal{F}_{X,m,\omega}(t, s) = [H(X, t) |t| \mathbf{N}(t; -\frac{3}{2} + i\omega, m)] B_{a,b}^{m,\omega}(s).$$

Both factors are absolutely integrable, so Fubini yields

$$\int_{\mathbb{C} \times \mathbb{C}} \mathcal{F}_{X,m,\omega}(t,s) d\mu_{\times}(t) d^2s = M_{a,b,s}(X;m,\omega) \beta_{a,b}(m,\omega). \quad (265)$$

Now fix $X \in E_1$, $m \in \mathbb{Z}$, and an arbitrary $\omega \in \mathbb{R}$. The homogeneous kernel admits the Schwinger representation Eq. (224); the absolute source–scale estimate permits the positive Schwinger scale integral to be moved outside. At each positive scale, the inner source/spatial integral is exactly the harmonic Gaussian moment $\mathcal{G}_{X,m,w}$ at the corresponding width $w > 0$, hence vanishes by Eq. (258). Therefore

$$M_{a,b,s}(X;m,\omega) \beta_{a,b}(m,\omega) = 0 \quad (266)$$

for every ω , on the same source-base set E_1 . No uncountable intersection of null sets occurs.

13.5 Evaluation and nonvanishing of the physical beta coefficient

Set

$$m_L = m + 2\delta j_{ab}, \quad m_R = m, \quad (267)$$

and

$$\lambda_L = \frac{1}{2} + i(\omega + 2\delta\nu_{ab}), \quad \lambda_R = \frac{1}{2} + i\omega. \quad (268)$$

Then

$$B_{a,b}^{m,\omega}(s) = K_{m_L,\lambda_L}(s) K_{m_R,\lambda_R}(1+s). \quad (269)$$

Let

$$p_L = p(m_L, \lambda_L), \quad p_R = p(m_R, \lambda_R). \quad (270)$$

Their real parts are

$$\operatorname{Re} p_L = \frac{|m_L|}{2} + \frac{3}{4}, \quad \operatorname{Re} p_R = \frac{|m_R|}{2} + \frac{3}{4}. \quad (271)$$

Inserting Eq. (224) twice gives

$$\begin{aligned} \beta_{a,b}(m,\omega) &= \frac{1}{\Gamma(p_L)\Gamma(p_R)} \int_0^\infty \int_0^\infty a^{p_L-1} b^{p_R-1} \\ &\quad \times \int_{\mathbb{C}} P_{m_L}(s) P_{m_R}(1+s) e^{-a|s|^2 - b|1+s|^2} d^2s da db. \end{aligned} \quad (272)$$

Absolute integrability follows from Eq. (263), so the order of integration is legitimate.

For same-sign m_L, m_R , completing the square in the spatial integral gives

$$\begin{aligned} &\int_{\mathbb{C}} P_{m_L}(s) P_{m_R}(1+s) e^{-a|s|^2 - b|1+s|^2} d^2s \\ &= \left(-\frac{b}{a+b}\right)^{|m_L|} \left(\frac{a}{a+b}\right)^{|m_R|} \frac{\pi}{a+b} e^{-ab/(a+b)}. \end{aligned} \quad (273)$$

The change of variables

$$r = a + b, \quad x = \frac{a}{a+b}, \quad da db = r dr dx$$

reduces the scale integral to one Euler Gamma integral and one Euler beta integral. The result is

$$\begin{aligned}\beta_{a,b}(m, \omega) &= (-1)^{|m_L|} \frac{\pi \Gamma(p_L + p_R - 1)}{\Gamma(p_L) \Gamma(p_R)} \\ &\times \frac{\Gamma(|m_R| + 1 - p_R) \Gamma(|m_L| + 1 - p_L)}{\Gamma(|m_R| + |m_L| + 2 - p_R - p_L)}.\end{aligned}\quad (274)$$

Every required Gamma argument lies in the proved open strip with positive real part, so no numerator or denominator factor vanishes or hits a pole.

For opposite-sign modes, physical helicity restricts the possibilities to

$$|m_L| + |m_R| = 8, \quad 1 \leq |m_L| \leq 7. \quad (275)$$

Let $d = |m_L|$, $8 - d = |m_R|$, and let p denote the combined Gamma parameter with

$$\operatorname{Re} p = \frac{11}{2}. \quad (276)$$

The mixed spatial Gaussian integral produces the finite diagonal moment sum

$$S_d(p) = \sum_{i=0}^d \mathbf{1}_{i \leq 8-d} (-1)^{d-i} \binom{d}{i} \binom{8-d}{i} \Gamma(i+1) \Gamma(p-i-1). \quad (277)$$

Repeated use of $\Gamma(z+1) = z\Gamma(z)$ factors out $\Gamma(p-5)$:

$$S_d(p) = \Gamma(p-5) P_d(p-5). \quad (278)$$

With $z = p-5$, $\operatorname{Re} z = 1/2$, and

$$P_d(z) = -z(z+1)(z+2)(z-4), \quad d = 1, 7, \quad (279)$$

$$P_d(z) = z(z+1)(z-3)(z-4), \quad d = 2, 6, \quad (280)$$

$$P_d(z) = -z(z-2)(z-3)(z-4), \quad d = 3, 5, \quad (281)$$

$$P_4(z) = (z-1)(z-2)(z-3)(z-4), \quad (282)$$

none of the polynomial roots can meet the vertical line $\operatorname{Re} z = 1/2$. Also $\Gamma(p-5) \neq 0$. Hence the opposite-sign coefficient is nonzero.

For the opposite-sign sector the complete coefficient is

$$\beta_{a,b}(m, \omega) = \frac{\pi}{\Gamma(p_L) \Gamma(p_R)} B(|m_R| + 1 - p_R, |m_L| + 1 - p_L) S_d(p_L + p_R), \quad (283)$$

with $d = |m_L|$. The Gamma denominators are nonzero in the physical strip, the Euler beta factor is nonzero because both of its arguments have positive real part, and S_d is nonzero by Eq. (278) and the quartic root test.

Combining the same-sign and opposite-sign sectors,

$$\boxed{\beta_{a,b}(m, \omega) \neq 0 \quad \forall a, b, m \in \mathbb{Z}, \omega \in \mathbb{R}.} \quad (284)$$

A detailed derivation of Eqs. (273)–(278) is collected in Appendix B.

Equations (266) and (284) now give

$$M_{a,b,s}(X; m, \omega) = 0 \quad \forall m \in \mathbb{Z}, \omega \in \mathbb{R} \quad (285)$$

for every $X \in E_1$.

13.6 Exact Mellin-to-Fourier identification

This step is where the Haar normalization matters. Using Eq. (202) and

$$t = e^{r+i\theta}, \quad d\mu_{\times}(t) = dr d\theta,$$

we have

$$\begin{aligned} |t| \mathbf{N}\left(t; -\frac{3}{2} + i\omega, m\right) \\ = e^r e^{(-3/2+i\omega)r} e^{im\theta} = e^{-r/2} e^{i(\omega r + m\theta)}. \end{aligned} \quad (286)$$

Therefore the Mellin coefficient is exactly

$$\begin{aligned} M_{a,b,s}(X; m, \omega) &= \int_{\mathbb{R}} \int_{-\pi}^{\pi} e^{-r/2} H_{a,b,s}(X, e^{r+i\theta}) e^{i(m\theta + \omega r)} d\theta dr \\ &= \int_{\mathbb{R}} \int_{-\pi}^{\pi} F_X(r, \theta) e^{i(m\theta + \omega r)} d\theta dr. \end{aligned} \quad (287)$$

This is the exact bridge; F_X is built from H , not directly from G . The relation to G is

$$F_X(r, \theta) = e^{3r/2} G_{a,b,s}(X, e^{r+i\theta})$$

from Eq. (205).

For every $X \in E_1$,

$$F_X \in L^1(\mathbb{R} \times (-\pi, \pi)). \quad (288)$$

The inverse-radius source estimate also gives the angular L^2 slice domain used in the Fourier-uniqueness theorem. Indeed, with $t = e^{r+i\theta}$,

$$\begin{aligned} |F_X(r, \theta)| &= e^{-r/2} e^{2r} \left| G_{a,b,s}(X, e^{r+i\theta}) \right| \\ &\leq e^{-r/2} e^{2r} \frac{C'_s}{e^r} = C'_s e^{r/2}. \end{aligned} \quad (289)$$

For each fixed r , the right side is independent of θ , and the angular interval has finite measure. Hence

$$F_X(r, \cdot) \in L^\infty(-\pi, \pi) \subset L^2(-\pi, \pi) \quad \forall r \in \mathbb{R}. \quad (290)$$

Equation (285) becomes

$$\int_{\mathbb{R}} \int_{-\pi}^{\pi} F_X(r, \theta) e^{i(m\theta + \omega r)} d\theta dr = 0 \quad \forall m \in \mathbb{Z}, \omega \in \mathbb{R}. \quad (291)$$

For fixed m , define

$$g_m(r) = \int_{-\pi}^{\pi} F_X(r, \theta) e^{im\theta} d\theta. \quad (292)$$

Fubini gives $g_m \in L^1(\mathbb{R})$, and Eq. (291) says its real-line Fourier transform vanishes identically. Hence

$$g_m(r) = 0$$

for almost every r . Since $m \in \mathbb{Z}$ is countable, one countable set of radii works for every angular mode. On that set, Eq. (290) supplies the angular L^2 domain, and completeness of the Fourier basis

$$\{e^{im\theta} : m \in \mathbb{Z}\}$$

in $L^2(-\pi, \pi)$ yields

$$F_X(r, \theta) = 0 \quad \text{for a.e. } (r, \theta). \quad (293)$$

Because $e^{-r/2} \neq 0$, Eq. (293) gives

$$H_{a,b,s}(X, t) = 0 \quad d\mu_{\times}\text{-a.e.}$$

for almost every source base point. Equation (202) and the fact that $|t|^2 > 0$ on $\mathbb{C}^\times$ imply

$$G_{a,b,s}(X, t) = 0 \quad d^2t\text{-a.e.}$$

for almost every source base point.

Finally, the center/chord chart is measure preserving. The unit input character in Eq. (89) never vanishes, and the factor $|t|^{-1}$ is nonzero off the null diagonal. Therefore vanishing of G recovers the original fixed-external-label source:

$$\iota_{a,b}s = 0. \quad (294)$$

We conclude:

$$\ker T_{\text{sh}}^{a,b} = \ker T_{\text{full}}^{a,b} = \{0\} \quad \text{on } \mathcal{R}_{\text{phys}}^{a,b}, \quad \forall a, b. \quad (295)$$

Equivalently, two sources in $\mathcal{R}_{\text{phys}}^{a,b}$ with identical cross-ratio celestial output are equal.

14 Main theorem and immediate consequences

The results separate a conditional retained-completion statement from an unconditional intrinsic injectivity theorem.

Retained completion. If

$$\|C_{\text{direct}}^{a,b}[s]\| \leq K \|\iota_{a,b}s\| \quad (s \in \mathcal{D}_{\text{pair}}) \quad (296)$$

holds uniformly in the external labels, then the retained core transform extends uniquely to

$$C_{\text{act}}^{a,b} : \mathcal{H}_{\text{src}} \rightarrow \mathcal{H}_{\text{ang}}^{\text{ret}},$$

the external family is strongly measurable, and the fiberwise maps assemble to the bounded physical transform

$$C_{\text{phys}} : \mathcal{H}_{\text{phys}}^{(2)} \longrightarrow L_{\text{ext}}^2(\mathcal{H}_{\text{ang}}^{\text{ret}}).$$

Its retained frame and observational quotient are those of Eqs. (67)–(71). No injectivity of this additional retained projection is asserted.

Intrinsic all-label injectivity. Without assuming Eq. (296),

$$\boxed{\forall a, b, \quad \ker T_{\text{sh}}^{a,b} = \ker T_{\text{full}}^{a,b} = \{0\} \quad \text{on } \mathcal{R}_{\text{phys}}^{a,b}.} \quad (297)$$

Equivalently,

$$T_{\text{sh}}^{a,b} \psi_1 = T_{\text{sh}}^{a,b} \psi_2 \implies \psi_1 = \psi_2 \quad (\psi_1, \psi_2 \in \mathcal{R}_{\text{phys}}^{a,b}).$$

The theorem retains unequal helicities, unequal input Mellin frequencies, every integer angular mode and every real Mellin frequency. The range $\mathcal{R}_{\text{phys}}^{a,b}$ is dense in $\mathcal{H}_{\text{src}}$, but no extension of injectivity to arbitrary completed source vectors is claimed.

Two immediate consequences are useful. The intrinsic Gram form

$$Q_{a,b}(\psi, \phi) = \langle T_{\text{sh}}^{a,b} \psi, T_{\text{sh}}^{a,b} \phi \rangle$$

is nondegenerate on $\mathcal{R}_{\text{phys}}^{a,b}$, and the intrinsic observational quotient

$$\mathcal{R}_{\text{phys}}^{a,b} / \ker T_{\text{sh}}^{a,b}$$

is canonically $\mathcal{R}_{\text{phys}}^{a,b}$ itself. These are injectivity statements, not stability statements: no lower frame bound, bounded inverse, or reconstruction estimate is proved.

15 Relation to Paper 5 and outlook

Paper 5 establishes that output shadow reflection carries a nontrivial unitary source transformation and that the canonical fixed-source transition is not fiberwise isometric on a positive-measure spectral set. Paper 6 leaves both facts intact. The source correction appears explicitly in the chord character and in the shifted mode/frequency parameters of the Mellin reduction, while Eq. (297) proves that these source-aware shifts do not create a kernel on $\mathcal{R}_{\text{phys}}^{a,b}$.

Thus the combined result is:

$$\boxed{\begin{aligned} &\text{the source correction is required by the physical kernel's Weyl law,} \\ &\text{fixed-source fiberwise isometry fails,} \\ &\text{and the corrected source-aware transform remains injective on } \mathcal{R}_{\text{phys}}^{a,b}. \end{aligned}} \quad (298)$$

Here “faithfulness” is used only as interpretive shorthand for this domain-specific injectivity statement; it does not denote a stronger representation-theoretic or stability property.

The paper stops before interaction. Later celestial dynamics would require an interaction on the physical pair carrier and a corresponding map at the angular-output level; no celestial OPE, collinear splitting amplitude, soft current, memory observable, or interacting Ward identity is constructed here. The noninteracting result handed forward is therefore a source-aware celestial transform with proved all-label injectivity on its intrinsic physical source range.

16 Conclusion

Paper 6 constructs the retained direct-integral physical transform conditionally on the explicit uniform core estimate and proves, independently of that completion hypothesis, that

$$\ker T_{\text{sh}}^{a,b} = \ker T_{\text{full}}^{a,b} = \{0\} \quad \text{on } \mathcal{R}_{\text{phys}}^{a,b}$$

for every physical external-label pair. Complete cross-ratio and full signed principal-series data therefore distinguish source vectors inside the intrinsic physical source range.

The result complements Paper 5 rather than undoing it: the source-aware correction remains required by the kernel's Weyl law, fixed-source fiberwise isometry remains false, and the corrected transform is nevertheless injective. No theorem is claimed for arbitrary vectors in completed $\mathcal{H}_{\text{src}}$, for the additional retained fundamental-sheet projection, or for stable inversion.

A Schwinger majorants for the Abel–Fresnel value

We record the analytic estimates underlying Proposition 12.7.

Let

$$n = n_{ab}, \quad \mu = \mu_{ab}, \quad h = \frac{|n|}{2}, \quad A = |Q| > 0, \quad B = |L|.$$

After the unit rotation $s = r_Q z$, the squared input character is represented by the compensated Schwinger slice

$$\mathfrak{S}_{n,\mu}(z, \tau) = \frac{\tau^{(|n|+2-i\mu)/2-1}}{\Gamma((|n|+2-i\mu)/2)} |z|^2 \Pi_n(z) e^{-\tau|z|^2}.$$

The real part of the Schwinger exponent is

$$\frac{|n|+2}{2} = h+1 > 0,$$

so the Gamma denominator is finite and nonzero.

At fixed $\tau > 0$, multiplication by

$$e^{i \operatorname{Re}(Az^2 + B_Q z)}, \quad B_Q = Lr_Q,$$

gives a Gaussian-polynomial moment of total degree $2h+2$. Completing the two-real-variable Gaussian square and evaluating the finite moments gives the bound

$$|\mathcal{M}_{a,b}(\tau; A, B_Q)| \leq C_{a,b}^{(0)} m_h(\tau; A, B),$$

where

$$m_h(\tau; A, B) = \begin{cases} \frac{(1 + B/\sqrt{A})^{2h+2}}{A^{h+2}} \tau^h \exp\left(-\frac{B^2}{8A^2}\tau\right), & 0 < \tau \leq A, \\ \frac{1}{\tau^2} \left(1 + \frac{B}{\sqrt{\tau}}\right)^{2h+2} \exp\left(-\frac{B^2}{8\tau}\right), & \tau \geq A. \end{cases} \quad (299)$$

As a representative Gaussian moment, in the equal-input-character mode $n = 0$ and with vanishing linear source,

$$\int_{\mathbb{C}} e^{-\tau|z|^2 + iA \operatorname{Re}(z^2)} d^2z = \frac{\pi}{\sqrt{\tau^2 + A^2}}, \quad (300)$$

and differentiation in τ gives the compensated second moment

$$\int_{\mathbb{C}} |z|^2 e^{-\tau|z|^2 + iA \operatorname{Re}(z^2)} d^2z = \frac{\pi\tau}{(\tau^2 + A^2)^{3/2}}. \quad (301)$$

The finite signed-mode moments used in the physical calculation are obtained from the same two-real-variable Gaussian generating function by finite differentiation and polynomial expansion.

The small-scale term is integrated by the substitution

$$u = \frac{B^2}{8A^2}\tau$$

when $B > 0$, with the $B = 0$ case obtained directly from the polynomial integral. The negative exponential absorbs every power of $B/\sqrt{A}$. The large-scale term is handled with

$$v = \frac{B^2}{8\tau},$$

and the same Gaussian absorption controls the polynomial factor. One obtains a constant C_h , depending only on h , such that

$$\int_0^\infty m_h(\tau; A, B) d\tau \leq \frac{C_h}{A} \quad (302)$$

for every $A > 0$ and $B \geq 0$.

The physical prefactor multiplying m_h is

$$C_{a,b}^{(0)} = \frac{\pi \mathfrak{c}_n}{|\Gamma((|n| + 2 - i\mu)/2)|},$$

where $\mathfrak{c}_n$ is the finite compensated Gaussian-moment constant for the signed mode n . Thus

$$|\mathcal{H}_{a,b}(Q, L)| \leq \frac{C_{a,b}^{(0)} C_h}{|Q|},$$

which is Eq. (218).

For $\delta > 0$ and fixed anchor A_0 , the rotated damped pairing has the Schwinger form

$$\mathcal{W}_\delta^{a,b}(A_0; Q, L) = (\chi_{a,b}^{\text{in}}(r_Q))^2 \int_0^\infty \mathcal{M}_{a,b}^{(\delta, A_0)}(\tau; A, B_Q) d\tau.$$

The finite Gaussian moment now contains the translated anchor $r_Q^{-1}A_0$; at fixed A_0 this changes only the finite linear/constant coefficients in the evaluated Gaussian moment. The radial denominator is shifted from τ to $\tau + \delta$. For each fixed $\tau > 0$,

$$\mathcal{M}_{a,b}^{(\delta, A_0)}(\tau; A, B_Q) \longrightarrow \mathcal{M}_{a,b}(\tau; A, B_Q) \quad (\delta \downarrow 0).$$

The same piecewise majorant in Eq. (299), enlarged by a fixed harmless constant for $0 < \delta \leq 1$, dominates the damped moments. Dominated convergence proves

$$\lim_{\delta \downarrow 0} \mathcal{W}_\delta^{a,b}(A_0; Q, L) = \mathcal{H}_{a,b}(Q, L).$$

The same uniform $1/|Q|$ estimate is what permits the later Gaussian spectral integral involving

$$K_\epsilon(\xi) \frac{J_{a,b}(\xi)}{|\xi|}$$

to be handled by ordinary dominated convergence; no unproved continuity of the undamped Fresnel value is needed.

B Derivation of the physical two-puncture beta coefficient

This appendix supplies the calculations used in Section 13.

B.1 Schwinger representation

For

$$K_{k,\lambda}(z) = |z|^{\lambda-2} e^{ik \arg z}, \quad p = \frac{|k| + 2 - \lambda}{2},$$

and $\text{Re } p > 0$,

$$\frac{1}{\Gamma(p)} \int_0^\infty a^{p-1} e^{-a|z|^2} da = |z|^{-2p}.$$

Multiplying by $|z|^{|k|} e^{ik \arg z}$ gives

$$|z|^{|k|-2p} e^{ik \arg z} = |z|^{\lambda-2} e^{ik \arg z},$$

because $|k| - 2p = \lambda - 2$. This proves Eq. (224).

B.2 Same-sign spatial Gaussian

Suppose $m_L, m_R \geq 0$; the nonpositive case is its complex conjugate. For $a, b > 0$, set

$$c = -\frac{b}{a+b}.$$

Then

$$a|s|^2 + b|1+s|^2 = (a+b)|s-c|^2 + \frac{ab}{a+b}.$$

After $u = s - c$,

$$\begin{aligned} & \int_{\mathbb{C}} s^{m_L} (1+s)^{m_R} e^{-a|s|^2 - b|1+s|^2} d^2 s \\ &= e^{-ab/(a+b)} \int_{\mathbb{C}} \left(u - \frac{b}{a+b}\right)^{m_L} \left(u + \frac{a}{a+b}\right)^{m_R} e^{-(a+b)|u|^2} d^2 u. \end{aligned} \quad (303)$$

Expanding both polynomials, rotational symmetry gives

$$\int_{\mathbb{C}} u^n e^{-(a+b)|u|^2} d^2 u = 0 \quad (n > 0),$$

so only the constant term survives:

$$\int_{\mathbb{C}} P_{m_L}(s) P_{m_R}(1+s) e^{-a|s|^2 - b|1+s|^2} d^2s = \left(-\frac{b}{a+b}\right)^{|m_L|} \left(\frac{a}{a+b}\right)^{|m_R|} \frac{\pi}{a+b} e^{-ab/(a+b)}.$$

Substitution into Eq. (272) yields an integral over $a, b > 0$. With

$$r = a + b, \quad x = \frac{a}{a+b}, \quad a = rx, \quad b = r(1-x),$$

the Jacobian is $da db = r dr dx$. Collecting powers gives

$$\int_0^1 x^{|m_R| - p_R} (1-x)^{|m_L| - p_L} dx$$

times

$$\int_0^\infty r^{p_L + p_R - 2} e^{-rx(1-x)} dr.$$

The r -integral is

$$\Gamma(p_L + p_R - 1) [x(1-x)]^{-(p_L + p_R - 1)},$$

and the remaining x -integral is the Euler beta integral. Simplifying produces Eq. (274).

B.3 Opposite-sign mixed moments

Assume $m_L \geq 0 \geq m_R$; the other orientation is obtained by conjugation. Put

$$d = |m_L|, \quad e = |m_R|.$$

After the same completion of the square,

$$s = u - \frac{b}{a+b}, \quad 1 + \bar{s} = \bar{u} + \frac{a}{a+b}.$$

Expanding gives

$$\sum_{i=0}^d \sum_{j=0}^e \binom{d}{i} \binom{e}{j} \left(-\frac{b}{a+b}\right)^{d-i} \left(\frac{a}{a+b}\right)^{e-j} u^i \bar{u}^j.$$

The centered Gaussian moment vanishes unless $i = j$, and

$$\int_{\mathbb{C}} u^i \bar{u}^i e^{-(a+b)|u|^2} d^2u = \frac{\pi \Gamma(i+1)}{(a+b)^{i+1}}.$$

Hence the spatial integral reduces to the single diagonal sum

$$e^{-ab/(a+b)} \sum_{i=0}^d \mathbf{1}_{i \leq e} \binom{d}{i} \binom{e}{i} \left(-\frac{b}{a+b}\right)^{d-i} \left(\frac{a}{a+b}\right)^{e-i} \frac{\pi \Gamma(i+1)}{(a+b)^{i+1}}. \quad (304)$$

After the positive-scale integrations, the physical opposite-sign sector has $d + e = 8$, and the remaining finite Gamma moment is Eq. (277). To verify the quartic factorization, write each $\Gamma(p - i - 1)$ as a polynomial multiple of $\Gamma(p - 5)$ using

$$\Gamma(w+1) = w\Gamma(w).$$

For example, $d = 1$, $e = 7$ gives

$$S_1(p) = -\Gamma(p-1) + 7\Gamma(p-2) = \Gamma(p-5)[-z(z+1)(z+2)(z-4)],$$

with $z = p - 5$. The cases $d = 2, 3, 4$ are obtained by the same recurrence; $d = 5, 6, 7$ follow from the symmetry $d \leftrightarrow 8 - d$. The resulting four polynomial families are exactly those displayed in Section 13.

Because $\operatorname{Re} z = 1/2$, none of their integral real roots is attained. Thus every opposite-sign coefficient is nonzero.

C Formal verification and reproducibility

A machine-checked Lean development accompanies the manuscript. The manuscript has been checked against current main at commit

27c28104be6f41d2d115d2ba4a46f0dcdb756d4d,

while the terminal Paper 6 all-label injectivity checkpoint is

0c78bfa3f1535a588228fa055dc448c03797ee9a.

The verification environment uses Lean

v4.28.0-rc1

and mathlib revision

5352afccd6866369be9de43f5b7ec47203555f44.

The formal development is organized into four parts:

Carrier and domain construction.

Direct-integral carriers, the ordered separable pair core, the Bose-projected physical algebraic core, affine source realization, source-domain estimates and the global incidence majorants.

Coherent transform and source descent.

Doubled-coherent absolute integrability, coherent L^2 membership, the independent global-flat construction, Gram comparison, source-kernel inclusion and quotient descent to $\mathcal{R}_{\text{phys}}^{a,b}$.

All-label uniqueness machinery.

Incidence/coarea identities, Gaussian and Schwinger estimates, quadratic testing, shifted Mellin factorization, same-sign and opposite-sign beta evaluations, and logarithmic-cylinder Fourier uniqueness.

Terminal theorem.

Recovery of the original source and all-label injectivity of $T_{\text{sh}}^{a,b}$ and $T_{\text{full}}^{a,b}$ on $\mathcal{R}_{\text{phys}}^{a,b}$.

The terminal audit reports no newly introduced axioms or unfinished proof placeholders in this theorem chain. Formal verification is supplementary evidence for derivability from the encoded definitions and hypotheses; the body and appendices are intended to make the domains, measure changes and decisive analytic steps independently inspectable.

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